

# **Weathering Change: Building Hyperlocal Climate Resilience in Germantown, Philadelphia**



Makayla Coleman, Natalie Schliekelman, Noah Kogut Levin, and Sarah Boruchov

A.B. Candidates in Environmental Studies, May 2026

ENVS 397: Senior Seminar in Environmental Studies

Submitted in partial fulfillment of the Environmental Studies Major

Professor Dylan Gauthier

May 9th, 2026

|                                                           |           |
|-----------------------------------------------------------|-----------|
| <b>Executive Summary:</b>                                 | <b>3</b>  |
| <b>Introduction:</b>                                      | <b>5</b>  |
| <b>Research Questions:</b>                                | <b>6</b>  |
| <b>Methods:</b>                                           | <b>7</b>  |
| <b>Definitions:</b>                                       | <b>9</b>  |
| <b>Approach:</b>                                          | <b>10</b> |
| <b>Limitations &amp; Ethical Considerations:</b>          | <b>10</b> |
| <b>Background:</b>                                        | <b>17</b> |
| <b>Climate Risk Overview:</b>                             | <b>22</b> |
| <b>Literature Review:</b>                                 | <b>27</b> |
| Environmental Justice                                     | 27        |
| Community Resilience Theories                             | 28        |
| Frameworks                                                | 32        |
| <b>Community Assets &amp; Organizations in Germantown</b> | <b>35</b> |
| <b>Vulnerability and Risk Assessment:</b>                 | <b>37</b> |
| Flooding                                                  | 38        |
| Air Pollution                                             | 43        |
| Heat                                                      | 49        |
| Food                                                      | 52        |
| <b>Climate Resilience Strategies:</b>                     | <b>53</b> |
| Flooding                                                  | 58        |
| Built Environment Solutions                               | 58        |
| Nature-Based Solutions                                    | 65        |
| Extreme Heat and Air Pollution                            | 69        |
| Food Systems                                              | 84        |
| <b>Community Engagement Strategy:</b>                     | <b>91</b> |
| <b>Conclusion:</b>                                        | <b>92</b> |
| <b>Appendices:</b>                                        | <b>95</b> |
| <b>Sources</b>                                            | <b>99</b> |

## **Executive Summary:**

This capstone project develops a hyperlocal climate resilience plan for Germantown, a historic neighborhood in Philadelphia facing overlapping environmental, infrastructural, and social vulnerabilities intensified by climate change. Germantown experiences disproportionate

exposure to flooding, extreme heat, air pollution, energy insecurity, and food system disruptions due to long-standing patterns of racial segregation, economic disinvestment, aging infrastructure, and environmental injustice. These vulnerabilities are compounded by climate change, which is increasing the frequency and severity of heat waves, heavy precipitation events, and public health risks throughout Philadelphia.

The goal of this project is to examine what an effective neighborhood-level climate resilience plan can look like when grounded in principles of environmental justice, community resilience, and hyperlocal planning. Rather than relying solely on broad citywide approaches, this project argues that climate resilience strategies must respond directly to the specific historical, social, environmental, and infrastructural realities of individual neighborhoods. Germantown serves as a critical case study because its climate risks are deeply interconnected with structural inequalities that shape residents' exposure, vulnerability, and adaptive capacity.

This project combines environmental justice scholarship, climate resilience theory, policy analysis, and vulnerability assessments to evaluate the major climate risks facing Germantown. Particular attention is given to flooding caused by buried waterways and aging sewer infrastructure, extreme heat intensified by the urban heat island effect, and poor air quality linked to traffic emissions and broader regional pollution patterns. The project also examines how these environmental hazards intersect with housing insecurity, food access, and public health disparities.

Using an equitable resilience framework, this plan proposes a series of community-centered resilience strategies designed to reduce environmental vulnerability while strengthening long-term neighborhood stability. Recommended strategies include tree canopy expansion, green infrastructure investments, cooling centers, weatherization programs, air

quality monitoring, public health outreach, anti-displacement protections, and participatory planning processes that center community knowledge and leadership. Importantly, the project emphasizes that climate resilience must extend beyond physical infrastructure to include social networks, public health systems, cultural preservation, and economic justice.

In addition to the written report, this project also includes plans for accessible community-facing educational materials developed in collaboration with local organizations such as The Water Shed. This dual-output approach reflects a commitment to democratizing climate knowledge and ensuring that residents are not passive recipients of information, but active participants in shaping climate adaptation efforts within their own community.

Ultimately, this capstone argues that equitable climate resilience planning requires a hyperlocal, justice-oriented, and community-driven approach. By centering Germantown's specific vulnerabilities, strengths, and histories, this project demonstrates how climate adaptation can simultaneously reduce environmental risks, strengthen community resilience, and advance long-term environmental justice.

**Introduction:**

The primary aim of this project is to develop a comprehensive, hyperlocal climate resilience plan that is specifically tailored to the unique environmental, social, and infrastructural conditions of Germantown. This plan will take the form of a detailed analytical report that synthesizes research and community insights. In addition, the project will produce community-facing pamphlets designed for collaboration with The Water Shed intended to share key findings in an accessible and engaging format for local residents.

This dual-output approach reflects a commitment to both academic rigor and public accessibility, recognizing that the democratization of knowledge is essential for fostering meaningful and sustained community resilience. Too often, climate research and policy discussions remain confined to academic institutions, government agencies, or technical experts, making critical information inaccessible to the communities most affected by environmental injustice. By translating complex climate data, policy recommendations, and public health information into clear, community-oriented language, this project seeks to bridge the gap between expert knowledge and everyday lived experience. Democratizing knowledge means ensuring that residents are not merely passive recipients of information, but active participants in understanding, shaping, and implementing climate resilience strategies within their own neighborhood.

This approach is especially important in Germantown, where long-standing racial and economic inequities have historically limited access to environmental resources, political power, and institutional support. Making information publicly accessible through pamphlets, visual materials, and collaborative outreach helps empower residents to advocate for healthier living conditions, engage with local policy discussions, and participate in community-based resilience

efforts. Furthermore, accessible educational materials can strengthen intergenerational knowledge-sharing, encourage collective action, and build trust between community organizations, researchers, and residents. In this way, the project positions knowledge itself as a tool of empowerment and climate justice rather than a resource reserved only for experts or policymakers. It can be used in conjunction with other community organizations' work toward climate resilience and social justice i.e. GREAT's Climate Resilience Plan with Drexel University.

The core objective of this project is to identify and assess key climate vulnerabilities specific to the Germantown context, analyze the underlying social and environmental inequities, propose actionable place-based solutions, and support community education and engagement. These objectives aim to advance a more resilient, informed, and empowered community in the face of climate change.

***Research Questions:***

The main question of our project is: What does an effective neighborhood-level climate resilience plan look like, and why is it necessary to create one for Germantown? Other questions we are considering are:

- What are the primary concerns of Germantown residents now? How do residents relate to climate issues?
- Why are hyperlocal plans more effective than city-wide approaches in certain contexts?  
Why does a hyperlocal plan matter to local residents?
- What is the importance of climate justice in climate resilience plans? How can we include justice in our plan? How can climate resilience planning address structural inequalities?  
What does a just resilience plan look like in practice?

- What has the city of Philadelphia done policy wise to help Germantown mitigate the impacts of climate change and flooding? Where are the gaps at the neighborhood level?
- Why are hyperlocal plans more effective than city-wide approaches in certain contexts?

### **Methods:**

This project employs an interdisciplinary and community-oriented methodological approach to develop a hyperlocal climate resilience plan for Germantown, Philadelphia. The methodology combines qualitative and analytical research strategies, including a comprehensive literature review, policy analysis, climate vulnerability assessment, spatial analysis, and community-informed research. Together, these methods were used to examine how environmental risks intersect with historical inequities, infrastructure, public health, and neighborhood-level vulnerability.

The project began with an extensive literature review focused on environmental justice, climate resilience theory, urban vulnerability, and equitable adaptation planning. Foundational scholarship from environmental justice theorists, resilience researchers, and climate adaptation frameworks was used to establish the conceptual foundation of the project. Particular attention was given to theories surrounding environmental racism, community resilience, hyperlocal planning, and green gentrification. This literature informed both the analytical framework and the development of proposed resilience strategies.

In addition to scholarly research, the project incorporated policy analysis and review of existing climate resilience initiatives within Philadelphia. Federal, state, and local climate adaptation policies were examined alongside existing environmental justice frameworks and resilience programs. This included analysis of initiatives related to flooding mitigation, urban heat reduction, air quality improvement, green infrastructure, and public health preparedness.

Reviewing existing policy frameworks helped identify strengths, gaps, and opportunities for more localized and equitable resilience planning within Germantown specifically.

A climate vulnerability and risk assessment was also conducted to evaluate Germantown's major environmental hazards and structural vulnerabilities. This assessment utilized demographic information, environmental justice screening tools such as EPA EJScreen, historical infrastructure research, climate projections, and publicly available spatial data. Particular focus was placed on flooding vulnerability associated with buried waterways and combined sewer systems, urban heat island impacts, air pollution exposure, and neighborhood-level environmental burdens. Spatial analysis and mapping tools were used to identify areas experiencing concentrated environmental risk and overlapping vulnerabilities.

The project also attempted to incorporate community-informed perspectives and local knowledge into the planning process. Community organizations, resilience initiatives, and grassroots environmental justice efforts within Germantown were researched to better understand neighborhood strengths, existing adaptive strategies, and community priorities. Organizations such as The Water Shed, Germantown Residents for Economic Alternatives Together (GREAT), and the Senior Environment Corps were examined as examples of local resilience-building and participatory environmental action.

However, the project encountered important limitations regarding direct community engagement. Due to an unforeseen accident involving The Water Shed and the constraints of a semester-long research timeline, opportunities for extensive participatory research and resident interviews were significantly limited. As a result, the project relied more heavily on secondary data, previous research, environmental assessments, and publicly available information than originally intended. While this limitation restricted the ability to incorporate direct resident

testimony, the project remains grounded in principles of environmental justice and recognizes the importance of community leadership, participatory planning, and lived experience within future resilience efforts.

Overall, this methodological approach seeks to balance academic research, policy evaluation, environmental analysis, and community-centered planning in order to produce a climate resilience framework that is both analytically rigorous and socially equitable. Rather than treating environmental hazards as isolated problems, the project examines the interconnected relationship between climate change, infrastructure, public health, housing, and systemic inequality within Germantown.

### ***Definitions:***

Community Resilience: a community's sustained ability to utilize available resources to respond to, withstand, and recover from adverse situations. It involves the interconnected networks that directly impact society at a community level, including socioeconomic, ecological, and built environments<sup>1</sup>.

Climate Resilience Plan: a strategic framework developed for communities, organizations, governments, etc. to anticipate, prepare for, and respond to the impacts of climate change. The plan outlines measures and actions for a region to increase resilience.

Key components<sup>2</sup>:

1. Climate Vulnerability Assessment - thorough assessment of the region's vulnerability to climate change impacts

---

<sup>1</sup> "What is Community Resilience?" n.d. Kansas Wesleyan University.  
<https://www.kwu.edu/crh/resources/what-is-community-resilience/>.

<sup>2</sup> "What Is A Climate Resiliency & Climate Adaption Plan?" n.d. Indigenous Aware.  
<https://www.indigenousaware.com/climate-resiliency-adaptation-plan>

2. Risk and Impact Analysis - analyzing potential risks and impacts of climate change
3. Adaptation/Climate Resilience Strategies - specific adaptation strategies based on identified challenges
4. Community Engagement - meaningful engagement with stakeholders and the community
5. Integration with Existing Plans (excluded from this plan)
6. Monitoring and Evaluation (excluded from this plan)
7. Funding and Implementation (excluded from this plan)

***Approach:***

This project adopts an integrative methodological approach that synthesizes multiple forms of analysis to develop an Equitable Climate Resilience Framework to the best of our ability. Importantly, this approach combines a top-down policy analysis with bottom-up community knowledge. By bridging institutional perspectives with insights grounded in community experience, the project seeks to produce a holistic and contextual understanding of climate resilience in Germantown. Rather than treating environmental hazards in isolation, this approach considers the compounding and overlapping nature of risks i.e. heat and flooding, alongside the social conditions that shape differential exposure and adaptive capacity.

In terms of action, the project places an emphasis on identifying solutions that are not just theoretical but feasible, scalable, and community-rooted in that they respond to the real concerns of Germantown's residents. Ultimately, this synthesis of methods is intended to advance a resilience planning process that is equitable and interdisciplinary to the lived realities of Germantown residents.

***Limitations & Ethical Considerations:***

A significant limitation encountered was the inability to work with The Water Shed and local Germantown residents due to an unforeseen accident at The Water Shed. This disruption substantially constrained opportunities for participatory research, limiting our capacity to incorporate firsthand community perspectives, lived experiences, and locally grounded knowledge into the project's development. Due to reduced community interactions, there is an inherent risk that our analysis may reflect an external interpretation of community needs rather than a report shaped with community input. In addition, this tragedy means our project has to rely on secondary data which, while valuable, limits how much current or nuanced data we utilize directly from Germantown. We are also limited by this being a semester-long project, limiting our interactions with Germantown, and the ability to fully flesh out our research on certain topics.

Despite these constraints, we have an ethical commitment to avoid extractive research, represent the community accurately, and support (non-replaceable) local voices. This is essential to stay true to the values of environmental justice, and for a hyperlocal approach to climate change resilience. We acknowledge the limitations of our perspective and recognize that Germantown's residents are indispensable. We position this project as a provisional and adaptive contribution which can be beneficial for future collaborations and dialogue with Germantown residents. This project is grounded in the iterative process, with the ability to build off our research and climate plans to further ground it in community needs.

In terms of ethics, a climate resilience framework for Germantown must recognize that climate change does not affect all communities equally. The impacts of flooding, extreme heat, air pollution, and environmental degradation are shaped by historical patterns of racial segregation, economic inequality, housing discrimination, and uneven public investment. As a

result, resilience planning cannot focus solely on environmental sustainability; it must also address social justice, public health, community stability, and equitable access to resources. Ethical climate planning requires decision-makers to consider not only how to reduce climate risks, but also who benefits from resilience investments, who may be harmed, and whose voices are included in the planning process.

Environmental justice should serve as a foundational principle within Germantown's climate resilience plan. Historically, many Philadelphia neighborhoods, including Germantown, have experienced uneven patterns of investment and environmental burden due to discriminatory housing policies, redlining, infrastructure placement, and economic disinvestment. Ethical planning requires acknowledging these historical inequities and ensuring that climate resilience resources are distributed fairly. A key ethical question is: who benefits first, and who may be left behind? Climate interventions such as tree planting, cooling centers, weatherization programs, and green infrastructure should prioritize areas experiencing the greatest heat exposure and air pollution burdens rather than concentrating investments in already advantaged areas. Equity-focused planning also requires ensuring that renters, not only homeowners, can access the benefits of resilience initiatives. Without intentional equity measures, resilience investments risk reinforcing existing inequalities rather than reducing them.

One of the central ethical tensions in urban climate resilience planning is the relationship between environmental improvement and displacement. Investments in parks, tree canopy expansion, and green infrastructure can unintentionally increase property values and accelerate gentrification, placing long-term residents at risk of displacement. This phenomenon, often referred to as green gentrification, raises difficult ethical questions about balancing climate adaptation with housing affordability and community preservation. Ethical resilience planning

must therefore pair environmental investments with anti-displacement protections, including affordable housing preservation, property tax relief programs, renter protections, and community land trust initiatives. Residents should also be meaningfully involved in decision-making processes from the beginning to ensure that neighborhood improvements reflect community needs and priorities rather than outside redevelopment interests. Climate resilience efforts should strengthen communities without forcing residents out of the neighborhoods they helped sustain.

Procedural justice refers to fairness in decision-making processes and emphasizes the importance of community participation in climate planning. A resilience plan is both more legitimate and more effective when residents directly shape policies and interventions affecting their lives. Ethical planning requires asking who is included at the decision-making table and whose voices are often excluded. Inclusive engagement should intentionally involve renters, seniors, youth, non-English speakers, disabled residents, and other groups that are frequently underrepresented in planning processes. Community engagement should move beyond symbolic “input sessions” that fail to influence outcomes. Instead, residents should have genuine decision-making power and ongoing opportunities for feedback and collaboration. Whenever possible, community members should also be compensated for their time, expertise, and labor, recognizing that lived experience is a valuable form of knowledge in climate resilience planning.

Climate change-related events like flooding or heat create disproportionate health risks for certain populations, making health equity a critical ethical consideration. Older adults, children, individuals with asthma or cardiovascular disease, outdoor workers, and low-income households often face greater vulnerability to climate-related health impacts. Ethical climate planning does not simply distribute resources equally; rather, it prioritizes those most at risk. For example, cooling centers are only truly equitable if they are physically accessible, connected to

reliable transportation, culturally welcoming, safe for vulnerable populations, and open during the times when residents need them most. Similarly, air quality interventions should prioritize schools, healthcare facilities, and neighborhoods with high asthma rates. Addressing health inequities requires integrating public health considerations into every stage of resilience planning.

Climate resilience planning often relies heavily on environmental models, heat maps, vulnerability indices, and air quality data. While these tools are valuable, they also raise ethical concerns regarding representation and accuracy. Outdated or overly generalized data can misrepresent neighborhood realities and overlook localized environmental conditions. Additionally, an overreliance on quantitative metrics can erase lived experiences and community knowledge. Ethical climate planning should therefore combine scientific data with resident perspectives and local observations. Community members often possess detailed knowledge about where temperatures feel hottest, where flooding occurs most frequently, or where pollution exposure is highest. Integrating both technical expertise and lived experience creates a more accurate and equitable planning process.

Transparency is essential for building public trust and ensuring accountability within climate resilience planning. Residents should clearly understand why specific decisions are made, how funding is allocated, and what tradeoffs are involved in different interventions. Ethical planning rejects “black box” decision-making processes in which technical experts or institutions make decisions without public explanation. Instead, resilience plans should openly communicate assumptions, uncertainties, and limitations while providing ongoing opportunities for community feedback and revision. Accountability mechanisms should also be established so

residents can monitor progress, evaluate outcomes, and advocate for adjustments when interventions are not meeting community needs.

Climate resilience planning carries ethical responsibilities not only to current residents but also to future generations. Decisions made today regarding infrastructure, housing, land use, and environmental investment will shape long-term community health and sustainability for decades to come. Ethical resilience planning therefore requires balancing immediate emergency responses with investments in long-term adaptation strategies such as tree canopy expansion, energy-efficient building retrofits, and sustainable infrastructure improvements. Planners must also avoid short-term solutions that shift environmental burdens into the future, such as relying on energy-intensive cooling systems without addressing underlying emissions or urban heat conditions. Intergenerational responsibility emphasizes the need to create resilient systems that remain sustainable over time.

Even well-intentioned climate interventions can produce unintended negative consequences if not carefully designed and monitored. Ethical planning requires anticipating and minimizing these potential harms. For example, tree planting programs without long-term maintenance funding may result in dead trees and wasted resources. Increased policing around cooling centers may unintentionally discourage vulnerable residents, particularly communities of color, from accessing services. Similarly, indoor air filtration programs could increase household energy costs for low-income families. Ethical resilience planning requires continuous evaluation of interventions, monitoring for unintended effects, and adapting policies over time to better meet community needs. Flexibility and responsiveness are essential components of ethical climate governance.

Moreover, Germantown possesses a rich cultural, historical, and architectural identity that should be respected throughout the climate resilience planning process. Ethical interventions should strengthen rather than erase neighborhood character and community traditions. Climate adaptation strategies must therefore be culturally sensitive and responsive to local priorities. One-size-fits-all approaches imported from other cities or neighborhoods may fail to recognize the unique social and historical context of Germantown. Preserving historic architecture, supporting community gathering spaces, and incorporating local cultural values into environmental design can help ensure that resilience efforts enhance rather than undermine neighborhood identity.

An additional ethical tension within climate resilience planning concerns the uneven distribution of responsibility for climate adaptation. Local communities such as Germantown are often expected to adapt to environmental problems they did not create. Residents facing extreme heat and pollution are frequently those with the fewest financial resources and the least political power. Ethical planning therefore requires advocating for support across multiple scales of government and industry. City, state, and federal agencies should provide substantial funding for resilience projects, while utilities and major polluters should also be held accountable for environmental harms. Local residents should not bear the financial or social burden of addressing systemic environmental injustices alone.

Climate change creates an ethical challenge between the urgency of immediate action and the need for inclusive, participatory planning. Extreme heat and air pollution pose immediate threats to public health, meaning some interventions must occur quickly to protect residents. However, meaningful community engagement and equitable planning processes require time, trust-building, and collaboration. Ethical climate resilience planning must balance these

competing demands by implementing urgent protective measures, such as emergency cooling access and public health outreach, while simultaneously building long-term participatory structures that allow residents to shape future decisions. Rapid action should not come at the expense of democratic inclusion and community empowerment.

Ultimately, an ethical climate resilience framework for Germantown must address both environmental and social systems simultaneously. Effective resilience planning should prioritize the most vulnerable populations, center community voices in decision-making, prevent displacement and green gentrification, promote transparency and accountability, and remain adaptable as environmental conditions and community needs evolve. By grounding climate adaptation in principles of justice, equity, and community self-determination, Germantown can pursue resilience strategies that not only reduce environmental risks but also strengthen the social fabric of the neighborhood.

### ***Background:***

Germantown, a historic neighborhood in Philadelphia, is one of the oldest continuously inhabited areas in the United States. Founded in 1683, Germantown's history has long been associated with early American democracy, abolition, religious freedom, and social reform. Although originally settled by European immigrants, Germantown today is a predominantly Black community, with over 75% of residents identifying as Black according to the 2010 Census. This demographic transformation was shaped largely by the Great Migration of the twentieth century, during which many Black families moved north in search of economic opportunity and escape from racial violence in the Jim Crow South. At the same time, discriminatory urban policies such as redlining, blockbusting, restrictive covenants, and segregation accelerated white

flight while concentrating Black residents in neighborhoods such as Germantown. These policies not only shaped demographic patterns, but also determined where public and private investment occurred, producing long-lasting inequalities in housing quality, infrastructure, healthcare access, environmental conditions, and economic opportunity.

Germantown has also played a significant role in Black liberation movements throughout American history. The neighborhood served as an important stop along the Underground Railroad and remains home to numerous Black heritage sites and historic institutions connected to abolition and civil rights organizing. Germantown is also associated with the 1688 Germantown Quaker Petition Against Slavery, widely recognized as one of the first formal protests against slavery in colonial America. Figure 1.a, a map created by Ardmore artist Jessie Husband, visually demonstrates Germantown's deep historical significance within broader Black liberation and freedom movements. This history reflects Germantown's longstanding role as a space of resistance, cultural preservation, activism, and community-building for Black residents in Philadelphia and beyond.



Figure 1. An artist rendition of the map of Germantown with historical facts and sites.

Despite this powerful historical and cultural legacy, Germantown has experienced decades of economic disinvestment and structural inequality. Racialized housing practices, uneven development, discriminatory lending, infrastructure neglect, and broader patterns of structural racism have contributed to persistent socioeconomic disparities and uneven environmental conditions that continue to shape residents' daily lives. Many of the neighborhood's homes and public infrastructure are aging, environmental resources are distributed unevenly, and residents face overlapping burdens connected to poverty, housing insecurity, pollution exposure, and climate vulnerability. These historic inequities have made Germantown particularly vulnerable to the impacts of climate change, including extreme heat, flooding, poor air quality, and energy insecurity.

For this reason, it is critically important to focus on hyperlocal areas such as Germantown when developing climate resilience plans. Climate change is often discussed at the global or

national scale, but its impacts are experienced most directly at the neighborhood level.

Hyperlocal climate planning recognizes that environmental risks are not distributed evenly across cities and that each neighborhood possesses unique social, historical, environmental, and infrastructural conditions that shape vulnerability. While citywide climate policies may establish broad goals, they often fail to fully capture the specific realities facing individual communities. Neighborhoods such as Germantown experience climate risks differently than wealthier or more recently developed areas of Philadelphia due to differences in housing quality, tree canopy coverage, infrastructure age, land use, income levels, health disparities, and access to environmental resources.

A hyperlocal focus is especially important because historical inequalities have produced uneven climate vulnerability within cities themselves. In Germantown, decades of disinvestment and racial segregation have contributed to aging sewer systems, reduced green space, limited environmental protections, and higher exposure to pollution and heat. As a result, residents may experience more severe flooding, higher indoor and outdoor temperatures, poorer air quality, and greater public health risks than residents in more affluent neighborhoods. These localized vulnerabilities cannot always be identified through generalized citywide climate data alone. Hyperlocal resilience planning allows policymakers, researchers, and community organizations to examine the specific environmental hazards affecting Germantown and develop targeted interventions that respond directly to neighborhood conditions.

Focusing on hyperlocal areas is also important because residents themselves possess valuable place-based knowledge that broader planning processes may overlook. Community members understand how environmental risks operate in their daily lives. Residents know which streets flood most severely during storms, which blocks become dangerously hot during summer

heat waves, where air quality feels worst, and which populations are most vulnerable during emergencies. This lived experience is essential for creating resilience strategies that are both effective and equitable. Hyperlocal planning therefore emphasizes community engagement and participatory decision-making, allowing residents to shape the policies and interventions affecting their neighborhoods rather than having solutions imposed externally.

Additionally, hyperlocal climate resilience planning helps address environmental justice concerns by recognizing that climate change intersects with systems of race, class, housing, and health inequality. Historically marginalized communities often face the highest environmental burdens while simultaneously possessing the fewest resources to recover from climate-related disasters. Without neighborhood-level analysis, resilience strategies may unintentionally reinforce existing inequities or overlook populations most in need of support. Hyperlocal planning allows for more equitable resource allocation by identifying where investments in tree canopy expansion, cooling infrastructure, flood mitigation, air quality improvements, housing retrofits, and emergency response systems are most urgently needed.

Neighborhood-level planning is also important because it strengthens community resilience beyond physical infrastructure alone. Climate resilience is not simply about protecting buildings or reducing environmental hazards; it also involves strengthening social networks, public health systems, local organizations, cultural institutions, and economic stability. Germantown possesses a strong sense of community identity, historical memory, and local activism that can serve as important assets in climate adaptation efforts. Schools, churches, community gardens, mutual aid networks, healthcare providers, and local nonprofits can all play key roles in emergency response, public education, resource distribution, and long-term

resilience-building. Hyperlocal planning recognizes that social cohesion and community trust are essential components of effective climate adaptation.

Furthermore, focusing on neighborhoods like Germantown allows climate resilience plans to address compounding and interconnected risks rather than treating environmental issues in isolation. Flooding, extreme heat, air pollution, housing insecurity, food insecurity, energy burden, and public health disparities are all interconnected systems that affect residents simultaneously. For example, extreme heat worsens air pollution, increases energy demand, and places stress on residents living in poorly insulated housing. Flooding can damage homes, worsen mold exposure, disrupt transportation, and increase economic hardship for already vulnerable households. Hyperlocal planning makes it possible to understand how these risks overlap within specific communities and to design integrated solutions that address multiple vulnerabilities at once.

Ultimately, Germantown represents an important case study for climate resilience planning because it illustrates how climate vulnerability is deeply connected to historical inequity, racial justice, and urban development patterns. A hyperlocal approach acknowledges that resilience cannot be achieved through universal or one-size-fits-all solutions. Instead, effective climate adaptation must be rooted in the unique environmental conditions, histories, cultures, and lived experiences of individual communities. By centering Germantown specifically, this climate resilience plan seeks to create targeted, community-driven, and equity-focused strategies that not only reduce environmental risks but also strengthen social wellbeing, preserve community identity, and promote long-term environmental justice.

### ***Climate Risk Overview:***

In recent years, climate change has become an increasingly urgent issue in Germantown. Residents face a wide range of environmental challenges affecting air, water, energy, food, and public health systems, many of which are intensified by historical inequalities that continue to compound present-day vulnerabilities. Long-standing patterns of redlining, housing discrimination, economic disinvestment, and uneven infrastructure development have contributed to disparities in access to environmental resources, green spaces, and neighborhood investment. These inequities shape how climate risks are experienced across Germantown, disproportionately burdening low-income residents, Black communities, seniors, renters, and other socially vulnerable populations. As a result, climate change in Germantown cannot be understood solely as an environmental issue; it is also deeply connected to questions of social equity, public health, housing stability, and economic justice.

One of the most pressing local concerns is urban flooding, with Germantown being one of the most flood-prone neighborhoods in Philadelphia. Historically, Germantown was built over buried waterways such as the Wingohocking Creek, which were converted into underground sewer systems during urban development. During heavy rainfall events, these aging sewer systems often become overwhelmed, causing infrastructure flooding in which water backs up into streets, homes, and basements. Flooding can occur rapidly and dangerously, posing serious threats to residents and, in some cases, contributing to fatalities. Climate change is worsening these risks because warmer atmospheric temperatures allow the air to hold more moisture, increasing the frequency and intensity of severe storms and heavy precipitation events. Flooding also creates secondary public health concerns, including mold exposure, water contamination, property damage, and housing instability, which disproportionately affect residents with fewer financial resources to recover.

Extreme heat is another major climate challenge in Germantown, particularly due to the urban heat island effect. Urban environments with large amounts of asphalt, concrete, rooftops, dense housing, and limited tree canopy absorb and retain heat, creating temperatures significantly higher than surrounding suburban or rural areas. In Germantown, neighborhoods with less green space and fewer environmental investments often experience the highest temperatures. The urban heat island effect disproportionately impacts low-income residents and Black communities, increasing the risk of heat exhaustion, heat stroke, dehydration, respiratory illness, and death during heat waves. High temperatures also contribute to increased energy demand and rising utility costs, placing additional financial strain on households already experiencing economic insecurity. Residents without access to air conditioning, cooling centers, or well-insulated housing are particularly vulnerable during extreme heat events.

Air pollution is another significant environmental and public health concern within Germantown. Poor air quality results from multiple interacting sources, including vehicle emissions, industrial activity, aging infrastructure, and climate-related factors such as rising temperatures and wildfire smoke. Increased heat contributes to the formation of ground-level ozone, which can worsen respiratory conditions and reduce overall air quality. Exposure to pollutants such as fine particulate matter (PM<sub>2.5</sub>) and ozone has been linked to asthma, cardiovascular disease, respiratory infections, and other chronic health conditions. These impacts are especially severe for children, older adults, and individuals with preexisting health conditions. Air pollution also reflects broader environmental justice concerns, as communities of color and lower-income neighborhoods are often located closer to pollution sources and experience higher cumulative environmental burdens.

Food systems and food access are also increasingly important climate resilience concerns in Germantown. Climate change threatens food security by disrupting supply chains, increasing food prices, damaging agricultural systems, and intensifying extreme weather events that affect food production and transportation. At the neighborhood level, residents already experiencing economic hardship may face additional barriers to accessing affordable, nutritious food during climate emergencies such as floods, heat waves, or power outages. Food insecurity can become especially severe for elderly residents, disabled individuals, and households without reliable transportation. Additionally, extreme heat can reduce the safety and quality of stored food during power outages while worsening conditions for workers involved in food production and distribution. Access to community gardens, urban agriculture initiatives, farmers markets, and local food distribution systems can strengthen neighborhood resilience by increasing local food access and reducing dependence on vulnerable external supply systems. In Germantown, community-based food systems can also support cultural preservation, public health, and environmental sustainability while helping residents adapt to climate-related disruptions.

Energy challenges further compound these environmental risks. Many homes in Germantown are older and less energy-efficient, leading to significant heat loss during winter months and excessive indoor heat retention during summer. These conditions increase household utility costs and reduce residents' ability to maintain safe indoor temperatures during extreme weather events. Residents are also vulnerable to power outages and grid stress, particularly during periods of extreme heat or severe storms when energy demand is highest. Limited access to energy-efficient upgrades, renewable energy technologies, and financial assistance programs further deepens existing inequalities. Energy insecurity intersects with public health, housing

quality, and economic stability, making it an important consideration in climate resilience planning.

Social equity is central to understanding how climate risks affect Germantown residents. Climate change does not impact all residents equally; rather, existing social and economic inequalities shape vulnerability, exposure, and recovery capacity. Residents facing poverty, housing insecurity, language barriers, health disparities, disability, or limited political representation often have fewer resources available to prepare for and recover from climate-related disasters. Renters may have limited control over housing conditions and energy efficiency improvements, while historically marginalized communities may experience reduced access to environmental investments and emergency services. Without intentional equity-focused planning, climate adaptation strategies can unintentionally reinforce displacement, exclusion, or green gentrification. As a result, climate resilience planning in Germantown must prioritize fairness, community participation, and protection for vulnerable populations.

Together, these overlapping challenges, including flooding, extreme heat, air pollution, food insecurity, energy insecurity, and social inequity, highlight the unequal distribution of environmental risks throughout Germantown. These vulnerabilities are not solely the result of climate change itself, but are deeply rooted in long-standing structural inequalities that continue to shape neighborhood conditions today. Consequently, these environmental burdens negatively affect residents' health, safety, housing stability, economic security, and overall quality of life.

This climate resilience plan seeks to assess the extent of these interconnected challenges while identifying strategies to reduce vulnerability, strengthen environmental systems, and promote a more equitable and sustainable future for Germantown. Importantly, the plan aims to advance environmental justice by protecting residents from immediate climate risks, addressing

compounding social and environmental inequalities, improving public education around climate change, strengthening community resilience, and ensuring that adaptation efforts benefit long-term residents rather than contribute to displacement. Through a combination of infrastructure investment, community engagement, public health interventions, and equity-focused policy solutions, Germantown can work toward a future that is both environmentally sustainable and socially just.

### **Literature Review:**

The subject of our research is on community resilience and the impacts of climate change within Germantown, Philadelphia, and how climate resilience plans can aid community members in adapting and responding to environmental stressors. We selected literature to provide a fundamental understanding of environmental justice and climate change, along with research on community resilience theories and planning frameworks to best structure our project. The basis for this literature review comes from a desire to mix academic research on environmental justice and climate resilience plans with lived experiences from those in Germantown.

### **Environmental Justice**

Environmental justice was coined by Robert Bullard, Paul Mohai, Robin Saha, and Beverly Wright in the 1980s to describe the distribution of environmental benefits and harms as a result of systemic and structural inequities. Environmental justice is the recognition of the unequal and unfair impacts of climate change. The EPA defines environmental justice as “the fair treatment and meaningful involvement of all people regardless of race, color, national origin, or income, with respect to the development, implementation, and enforcement of environmental laws, regulations, and policies” (*Term: Environmental Justice*, 2013). Marginalized communities (low-wealth, rural, immigrant/refugee, indigenous, racial/ethnic groups) are disproportionately

harmed by environmental hazards and the impacts of climate change. Environmental hazards include the heightened exposure to pollution, toxic waste, oil refineries and urbanization. Environmental injustice is also experienced through limited access to environmental services and the loss of land and resource rights (*Center for Sustainable Systems*, n.d.).

Robert Bullard writes about the environmental justice framework as it attempts to prevent environmental threats before they occur, incorporating facets of other social movements that work to eliminate harmful practices. The framework strives to uncover underlying assumptions that contribute to and produce unequal environmental protection. He distills equity into three main categories: procedural, social, and geographic. Procedural is the extent that governing rules, regulations, evaluation criteria, and enforcement are applied fairly and uniformly without discrimination. Social equity is the role sociological factors from race, ethnicity, culture, class, lifestyles, political power, etc. have on environmental decision making. Lastly, geographic equity is the location and spatial configurations of communities and their proximities to environmental hazards. Environmental racism is the systemic disadvantage (through policy, practices, or directives) of individuals, groups, or communities based on race, combining public policies and industry practices to provide benefits for whites, while shifting harms onto people of color. Bullard notes how environmental decision makings and policies reflect the power arrangements of the dominant society and institution, highlighting the role environmental racism has in dictating access to environmental services or exposure to environmental and health risks.

### *Community Resilience Theories*

In working to mitigate impacts of climate change and environmental justice, a major component to adaptation and resilience strategies has focused on building local communities and networks' capacity to anticipate, prepare for, respond to, and recover from environmental

hazards and disasters. Effectively combating climate risks and integrating the community involve a focus on equity, supporting vulnerable populations, and ensuring long-term sustainability and livelihoods. Building community resilience cultivates safe and sustainable communities that can thrive, despite environmental issues, enhancing the populations' physical and psychological health, and producing long-term economic, social, and environmental stability.

A resilient community encompasses a variety of components, including the ability to anticipate problems, opportunities and potentials for surprises; reduce vulnerabilities in development paths, socioeconomic conditions, and sensitivities to threats; respond effectively and fairly to emergencies; and recover rapidly, better, safer, and fairer from climate disasters (Wilbanks 2008 cited by Morrow 2008). Effective community resilience and networks are driven by bottom-up efforts, addressing the main needs of a community, with local residents acting as leaders, thinkers, and researchers, rather than top-down technocratic solutions that can often ignore or overlook underlying factors and causes to environmental issues, such as flooding, urban heat, pollution, biodiversity loss, health risks etc. Top-down solutions also have the potential to worsen inequities if they are not grounded in social practices and values focused on capacity building and knowledge co-production (Knox-Hayes et al. 2025). Community resilience empowers local networks to prepare for, withstand, and recover from environmental hazards by improving internal resources and social connections; this shift from top-down aid to bottom-up action improves self-sufficiency and crisis response, efficiently utilizing local knowledge and getting to the root of systemic inequities. Morrow notes how community resilience is largely defined by social and economic forces, ties to the economic and political circumstances of the community, along with the strength of social institutions and networks. Marginalized communities that rely on resilience the most, may have the hardest time achieving it due to large

disparities in the distribution of resources. Community involvement is essential for community resilience, relying on grassroots advocacy to build strong community social structures, and for public officials to earn the community's trust.

Integral to equitable climate solutions is “thinking globally, acting locally”, where the worldwide impacts of decisions are considered in smaller, manageable actions within a community. Local governance, and community resilience can empower individuals to fight climate change and promote sustainability through grassroots initiatives, rather than relying on top-down funds or policies. The impacts of climate change are ever-increasing, disproportionately impacting communities of color and low-income communities. The federal government is also actively prioritizing climate action, leading local governments with the challenge of enhancing community resilience and mitigating the devastating impacts of climate change. Best et al. find four challenge areas in local-scale planning and finance for equitable climate resilience:

- capacity and governance,
- adaptation and recovery,
- funding mechanisms,
- and data and methodological needs.

Many local governments lack resources to facilitate better decision-making, limited by financial and human support constraints. Improving capacity and governance involves learning how to empower communities for adaptive learning and governments, and how to adapt to fast-changing environmental conditions. Adaptation and recovery must be treated as separate goals, especially when applying for federal funding. These community definitions will have implications in grant eligibility, how projects are designed, desired outcomes, and the

communities that are considered. Funding mechanisms can be a major block to addressing community environmental needs. Improved transparency and sharing of data about investments in climate resilience actions can allow governments to direct funds in a more effective and informed manner, rather than relying on tax bases and existing funding measures. Lastly, data and methodological needs often have large gaps in research and information on environmental disasters, damages, livelihood practices, social networks, and culture. Focus must be not only on improving the measurement and reporting of disaster and climate change damages, but also how to better translate scientific findings into action strategies and practical information that local policymakers and communities can access. Focusing on these four key areas allows us to think beyond a singular environmental disaster, and rather the cumulative impacts of climate change, especially those within marginalized communities. By bridging community resilience and local planning, environmental justice can be effectively addressed, demonstrating the need for collaboration between local government and community members.

The most detrimental, yet ignored, impacts of climate change are on marginalized or environmental justice communities, who suffer a disproportionate burden of environmental hazards, pollution, and climate impacts. Oftentimes, policies are focused on large-scale projects (such as state or country wide) of specific environmental events (like a major disaster). But, there are communities throughout the world facing residual impacts of climate change. These may include: health risks from increased pollution; housing damage from flooding; lack of resources; housing precarity; lack of insurance; and more. In his chapter on reimagining environmental justice in the anthropocene, Schlosberg analyzes 2 different forms of disruption: immediate shocks and slow violence. Drawing upon Rob Nixon's theory of slow violence, a gradual and concealed form of harm often from environmental degradation, climate change, or systemic injustice that

disproportionately impacts marginalized communities, Schlosberg argues that community engagement in adaptation planning must identify and address the slow-moving and long-term disruptions, rather than just bouncing back from an immediate shock. Thus, community resilience is a social and democratic process driven by place-based attachment, participatory governance, and the ability to live through immediate shocks and slow violence. Resilient communities can become even stronger through adaptation planning that emphasizes material and emotional place attachments, or the bond individuals have with their environments. Environmental degradation and climate change severs one's attachment to place, generating injustices and limiting a community's ability to adapt.

### *Frameworks*

Climate resilience and adaptation planning follow a myriad of frameworks to promote effective and equitable resilience and recovery planning, such as the Sustainable Livelihoods Approach, the capabilities approach, the Household Livelihood Resilience Approach, the Disaster Resilience Integrated Framework for Transformation, and more. This literature review and subsequent climate resilience plan draw upon the equitable resilience framework due to its commitment to equitable responses for long-term environmental solutions and transformations. The Equitable Resilience Framework (ERF) attempts to make climate change planning more equitable, just, and effective to generate long-term social, cultural, economic, and environmental transformations. Knox-Hayes et al. developed the ERF to address the historical shortcomings of past frameworks by reconceptualizing the links between resilience and equity within communities. The ERF facilitates just solutions whilst empowering overburdened environmental justice communities. The ERF emphasizes three key domains of knowledge convergence: communicating climate science and impacts, incorporation of community-based data into climate

science and policy, and enhancing consultation in the planning, design, and policy process through community recognition. All of these domains emphasize increased civic participation, such as increasing representation, participatory/citizen science measures, community engagement and stakeholders, etc. The overarching goal is to transform social and economic systems to benefit all of society, rather than perpetuating patterns of exploitation and injustice. The ERF ensures climate adaptation enhances community well-being and equity, rather than just protecting property and economic gains.

### *Green Gentrification*

In 2017, Kenneth Alan Gould and Tammy L. Lewis published *Green Gentrification: Urban Sustainability and the Struggle for Environmental Justice*, one of the foundational texts examining the relationship between urban greening initiatives and gentrification. Their work challenges the assumption that sustainability projects are inherently equitable by demonstrating how environmental improvements can unintentionally contribute to displacement in historically marginalized communities. Gould and Lewis argue that investments in parks, green infrastructure, bike lanes, waterfront redevelopment, and environmental restoration often increase property values and attract wealthier residents, while long-term low-income residents, particularly Black and Latinx communities, face rising rents, evictions, and cultural displacement. Rather than viewing sustainability as universally beneficial, the authors emphasize that environmental improvements exist within broader systems of capitalism, real estate speculation, and racial inequality. Their work helped establish the concept of “green gentrification,” which refers to the process by which environmental revitalization becomes intertwined with urban redevelopment and displacement pressures.

This scholarship was later expanded upon by Alessandro Rigolon and colleagues, who further explored the cyclical and interconnected relationship between greening and gentrification. Rigolon et al. developed a framework that demonstrates how greening initiatives and gentrification can occur in multiple sequences. In some cases, greening projects may precede gentrification by making neighborhoods more attractive to developers and higher-income residents. In other cases, gentrification may occur first, followed by increased environmental investments as wealthier residents demand more amenities and public green spaces. Their model emphasizes that there is no single or linear pathway connecting urban greening and displacement; instead, the two processes are deeply intertwined and mutually reinforcing.

Another important theory supporting the relationship between greening initiatives and gentrification is the “green space paradox.” In 2014, Jennifer Wolch, Jason Byrne, and Joshua Newell introduced this concept in their article, *Urban Green Space, Public Health, and Environmental Justice: The Challenge of Making Cities “Just Green Enough.”* The authors argue that while green spaces are often created to address environmental injustice and improve public health in underserved communities, these same improvements can unintentionally trigger rising property values and housing costs. As neighborhoods become greener, healthier, and more aesthetically desirable, they may also become more attractive to developers and higher-income residents, initiating processes associated with early-stage gentrification. This creates a paradox in which environmental improvements intended to benefit marginalized communities may ultimately contribute to their displacement.

Wolch and her colleagues propose the concept of creating neighborhoods that are “just green enough,” meaning that cities should pursue environmental improvements while intentionally limiting speculative redevelopment and displacement pressures. Their work

highlights the need for anti-displacement policies, affordable housing protections, and community-led planning alongside sustainability initiatives. This theory was later expanded upon by Hamil Pearsall and Jillian Eller in their study, *Locating the Green Space Paradox: A Study of Gentrification and Public Green Space Accessibility in Philadelphia, Pennsylvania*. Pearsall and Eller specifically examine Philadelphia as a case study, analyzing how increased access to green spaces often overlaps with patterns of neighborhood change and demographic shifts. Their research demonstrates that environmental amenities are not distributed equally and that the benefits of urban greening are frequently accompanied by social and economic consequences for long-term residents.

Together, these theories and case studies demonstrate that urban greening cannot be understood solely as an environmental or aesthetic issue. Instead, greening initiatives exist within broader political, economic, and racial systems that shape who benefits from environmental improvements and who bears the burden of displacement.

### **Community Assets & Organizations in Germantown**

The Philadelphia Climate Justice Collective (PCJC), facilitated by The Environmental Collaboratory (TEC) at Drexel University published a report in 2024 to guide Philadelphia's climate change resilience and develop a local, environmentally-just climate transition plan. By collaborating with community members and leaders with direct experience with climate challenges, the report presented community-driven, equity-focused recommendations to improve environmental justice in the city. Three main thematic categories emerged in the report: addressing systemic conditions, integrated climate justice approach, and community leadership building. The report emphasizes the need to address both short and long-term climate issues, and engage with a variety of community members. By centering community resilience and

participation in these climate recommendations, these solutions prioritize a community-driven environmentally just climate transition plan, starting with grassroots advocacy and community organizations.

Germantown boasts a multitude of community organizations, working on environmental and climate justice, along with community participation. Germantown Residents for Economic Alternatives Together (GREAT) is a community collective focusing on utilizing resources to “grow food, share tools, support each other in times of need, organize around issues of housing justice, and strengthen our holistic health and well-being”. The grassroots organization is rooted in social justice, with learning circles, emergency predation, housing and development, food and seeds, mutual aid funds, and more, to promote community engagement and resilience. Many of their tactics replicate those described above, from local actions, place-based attachments, sharing knowledge and resources, and defining solutions to their own challenges, rather than waiting for governmental assistance or aid.

The Water Shed is a community hub in Germantown, operated in partnership with the Philadelphia Water Department, Mural Arts Philadelphia, and the Academy of Natural Sciences of Drexel University. The Water Shed focuses on flooding issues in Germantown, empowering flood resilience through community-driven art, resources, education, and science. The hub has become an integral part to the neighborhood, bringing together residents, artists, and scientists to create participatory community events and art projects, allowing residents to share their personal experiences with flooding and its impacts. Through creative community engagement and collaboration from a variety of fields, The Water Shed exemplifies the importance of community resilience tactics and supporting vulnerable populations to ensure long-term sustainability and livelihoods.

The Senior Environment Corps (SEC) is a volunteer program based in Germantown that connects seniors with meaningful opportunities to contribute to environmental conservation and science, while also fostering social connections and community. SEC is a prime example of citizen science in Germantown, allowing people to learn more about climate change and local environmental issues, all while promoting community outreach and resilience. This community group is important to highlight due to their action and presence within the community, their valuable work of citizen science, and how they could contribute to enacting a climate resilience plan.

### **Vulnerability and Risk Assessment:**

In 2001, the Intergovernmental Panel on Climate Change (IPCC) introduced a climate vulnerability framework, which defines it by three factors: sensitivity, adaptive capacity, and exposure.<sup>3</sup> IPCC defines vulnerability as “the degree to which a system will respond to a given change in climate, including beneficial and harmful effects,” adaptive capacity as “the degree to which adjustments in practices, processes, or structures can moderate or offset the potential for damage or take advantage of opportunities created by a given change in climate” and exposure as the degree to which the system encounters climate hazard events. This vulnerability assessment will consider these three factors to evaluate the structural vulnerabilities in Germantown, as well as the vulnerability of key Germantown assets to the impacts of climate change.

With climate change increasingly affecting systems, from weather to ecological to economic, vulnerabilities are becoming increasingly erratic. The margins of vulnerability have widened; a house that once sat just above the 100-year flood line may now find themselves

---

<sup>3</sup> IPCC, 2001: Climate Change 2001: The Scientific Basis. Contribution of Working Group I to the Third Assessment Report of the Intergovernmental Panel on Climate Change [Houghton, J.T., Y. Ding, D.J. Griggs, M. Noguer, P.J. van der Linden, X. Dai, K. Maskell, and C.A. Johnson (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 881pp.

below it when 500-year floods start occurring every decade, or the line past which heat becomes dangerous for elderly people is crossed more and more frequently over a summer. These next sections will discuss four areas of climate-related vulnerability for Germantown: flooding, air pollution, heat, and food systems. Through an environmental justice lens, each of these vulnerabilities are intertwined, affecting and exacerbating each other, especially when in combination with socioeconomic or minority vulnerabilities.

### ***Flooding***

Compounding factors make Germantown particularly vulnerable to climate risks. Long-term infrastructural issues have made Germantown sensitive to flooding, heat issues, and other climate vulnerabilities, while inhibiting Germantown's adaptive capacity, as the city's ability to take preventative measures is limited. Building flood resiliency, therefore, is integral for Germantown's flood readiness and response. A flood resilient city is one that has the ability to "resist, absorb, accommodate to and recover from the effects of a flood hazard in a timely and efficient manner, including through the preservation and restoration of its essential basic structures and functions"<sup>4</sup>. Flood resilient cities are impacted less by extreme flood events, and have a better ability to bounce back when they are hit.<sup>5</sup> Flood resiliency has increasingly become a focus on the global scale when it comes to flooding, as it offers a more interconnected view of mitigation and resistance; flood resilience incorporates four capacities: *avoid* damage through structural measures, *reduce* damage when flooding events occur, *recover* quickly to an equivalent state, and to *adapt* to an unreliable future. Looking at long-term flooding and the long-term onset

---

<sup>4</sup> United Nations Office for Disaster Risk Reduction (UNDRR). 2017. The Sendai Framework Terminology on Disaster Risk Reduction. "Resilience". Accessed 15 December 2025. <https://www.undrr.org/terminology/resilience>.

<sup>5</sup> Hammond, M. J., Chen, A. S., Djordjević, S., Butler, D., & Mark, O. (2015). Urban flood impact assessment: A state-of-the-art review. *Urban Water Journal*, 12(1), 14–29. <https://doi.org/10.1080/1573062X.2013.857421>

impacts of climate change, the slower dynamics of resilience will ensure that communities can become more flood-prepared over time, rather than reacting to flooding events as they happen. Resilience aims to build secure foundations and systems for flood mitigation and response, moving beyond only addressing the symptoms of flooding. Governments must foster trust and collective responsibility in building community flood resilience. Building flood resiliency for Germantown means reducing the impact of structural vulnerabilities while increasing the capacity for community support.

Germantown's infrastructure and topography lead to unusually high levels of pluvial flooding, which occurs when short, intense rainfall overflows the drainage system and surpasses the rate at which water can infiltrate into the ground. Germantown sits within a unique 'bowl-shaped' topography, giving it a higher sensitivity to flooding, as water from surrounding areas flows downhill and pools in Germantown. However, the primary contributor to Germantown's uniquely high vulnerability to pluvial flooding events are the buried historic streams. In the early 20th century, the Philadelphia Water Department undertook a project to divert the many streams of Germantown into pipes, the largest Philadelphia infrastructure project ever undertaken at the time.<sup>6</sup> The project was intended to prevent "dirty" urban streams from reaching the Schuylkill, where they would contaminate the clean river and a primary source of Philadelphia's drinking water. The engineers built the pipes into a combined system, where wastewater from houses and businesses would filter into the same pipes that the stormwater entered through intake grates on the streets. However, decades later, this creates a distinct climate vulnerability: when flooding events occur and the pipe system becomes overfilled, stormwater backs up out of not only intake grates, but seeping directly into people's basements. Runoff can

---

<sup>6</sup> Hopf, Amy. "Germantown Flooding: Community Connections to the Past, Present, & Future." @PhillyH2O Blog, July 28, 2021. <https://water.phila.gov/blog/germantown-flooding-task-force/>.

quickly overwhelm sewers, and with the already unusually high water table, flooding issues can compound quickly. The dense urban design of Germantown also means that it is highly susceptible to flash flooding during intense rainfall.

The rising temperatures and shifting rainfall patterns of climate change are increasing the intensity and frequency of floods. Average annual precipitation in Pennsylvania has increased 5-10% over the last century, and precipitation from extremely heavy storms in the Northeast has increased 70% since 1958.<sup>7</sup> Higher tides are expected along the Delaware River, with the tidal portion of the river expected to rise one to four feet in the next century<sup>8</sup>. Although this does not have as direct an impact on Germantown as on the Philadelphia neighborhoods closer to the Delaware River, a rise in the water table will mean that Germantown's water absorption capacity is lowered, and the floodwaters inch closer to the surface. Increased storms and flooding also means increased runoff, and increased pollutants running from the land into the water, decreasing water quality.

The combined sewer system that runs through Germantown means that every flood is also a potential human health risk. Combined sewer systems, which about 60% of Philadelphia is served by, transport wastewater from homes and buildings in the same drainage infrastructure as runoff from storms. During dry weather, the system transports all polluted water to treatment plants, but when a storm surge occurs, treatment plants can quickly reach capacity and combined sewers can overflow, releasing a mix of sewage and stormwater into local rivers and creeks, and in Germantown, basements and the streets.<sup>9</sup> A study from the Boston University School of Public Health found that in certain scenarios the risk of acute gastrointestinal illness increases by up to

---

<sup>7</sup> United States Environmental Protection Agency. "What Climate Change Means for Pennsylvania." <https://www.epa.gov/>, August 2016.

<sup>8</sup> U.S. EPA, "What Climate Change Means for Pennsylvania"

<sup>9</sup> Water.Phila.Gov. "CSOcast." Accessed May 9, 2026. <https://water.phila.gov/maps/csocast/>.

62% in the days following large-volume combined sewer overflows.<sup>10</sup> As extreme rainfall events occur more frequently, frequent combined sewer overflows may put residents' health at risk even further. The largest sewage releases occur when precipitation is most intense, a figure that is particularly concerning because heavy rainfall events have already been shown to have increased drastically in the Northeast. Waterways where a combined sewer outfall is actively overflowing are unsafe for any type of recreation or interaction, and may remain unsafe for a period of up to 72 hours following storm events. According to a study by PennEnvironment, sewage flows into most of Philadelphia's local waterways 65 times or more per year.<sup>11</sup> The Philadelphia Water Department estimates that in a typical year, 13.1 billion gallons of raw sewage mixed with polluted stormwater are dumped through combined sewer outfalls into local waterways.<sup>12</sup>

---

<sup>10</sup> Haley, Beth M., Yuantong Sun, Jyotsna S. Jagai, et al. "Association between Combined Sewer Overflow Events and Gastrointestinal Illness in Massachusetts Municipalities with and without River-Sourced Drinking Water, 2014–2019." *Environmental Health Perspectives* 132, no. 5 (2024). <https://doi.org/10.1289/ehp14213>.

<sup>11</sup> PennEnvironment Research & Policy Center. "Sewage Pollution in the Philadelphia-Camden Region." October 27, 2025. <https://environmentamerica.org/pennsylvania/center/resources/sewage-pollution-in-the-philadelphia-camden-region/>

<sup>12</sup> Sell, Lauren. "Response to PennEnvironment Sewage Pollution Report." *Philadelphia Water Department*, March 4, 2026. <https://water.phila.gov/drops/response-to-pennenvironment-sewage-pollution-report/>.

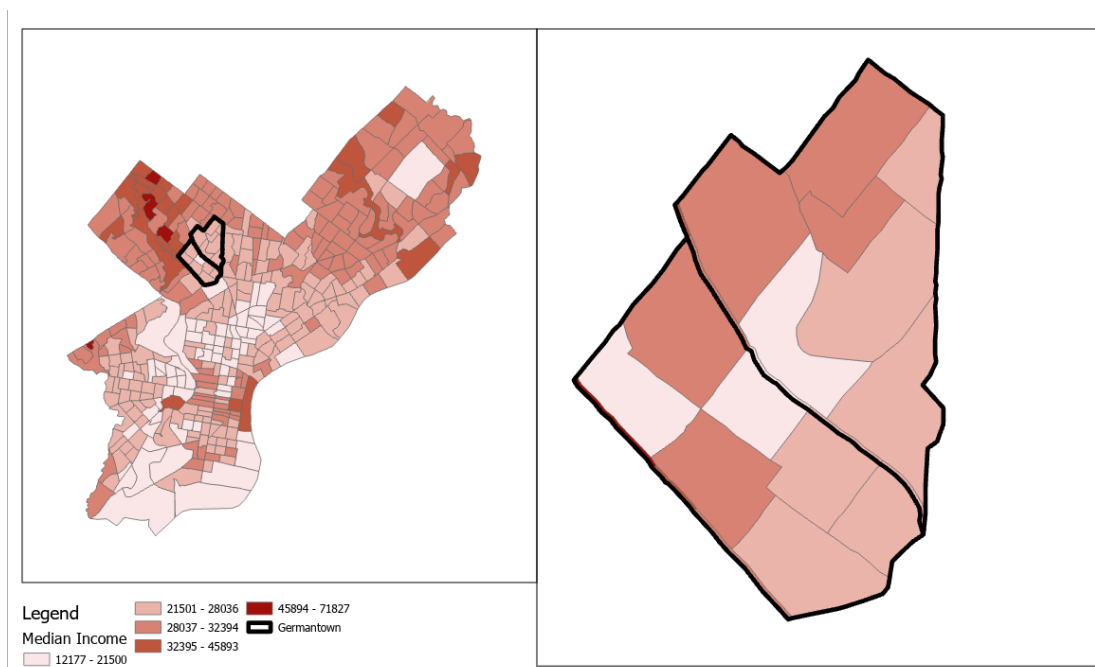


Figure 5: Median Income in Philadelphia

Flooding damage is a significant financial and physical burden on Germantown residents. Low- and moderate-income (LMI) communities experience a disproportionate impact from extreme weather and flooding, both in terms of exposure and in sensitivity/adaptive capacity. These differences in severity can be the result of socio-economic, political, and urban inequalities across households, leading to differences in vulnerabilities. Discrepancies between flooding protection infrastructure provided to higher income and lower income areas can also be a large factor. Overall, in flood-prone areas, lower-income households are often much more severely affected than high-income households because they tend to live on land and in homes that are more vulnerable to disasters. Lower-income homeowners often do not have the resources to undertake pre-storm mitigation practices. When flooding events occur, they are much more reliant on the resources that the National Flood Insurance Program provides for flood insurance to repair their homes, as compared to higher-income individuals who may have more fallback resources.

With exposure to frequent flooding, Germantown's capacity for response is pushed to its limit. The aging infrastructure throughout the community is not enough to deal with all of the flooding issues. With the issue of the covered streams at the heart of Germantown flooding, an issue that is prohibitively expensive and complicated to assuage, structural adaptations are limited to mitigative strategies rather than truly preventative ones, permanently lowering Germantown's adaptive capacity. This year, 33.7% of properties in Germantown have a risk of flooding, a statistic which is predicted to increase to 35.9% within the next 30 years.<sup>13</sup> Flooding, via all three characteristics, is a distinct vulnerability for Germantown residents and infrastructure, a situation which will not likely have an easy solution. Increasing flood resilience in Germantown, via infrastructure, policy, and community-building will improve Germantown residents' quality of life in the present, and will mean that Germantown is better equipped to handle the rainstorms and floods of the future.

### ***Air Pollution***

Air pollution refers to the presence of harmful substances in the atmosphere that can negatively affect human health, ecosystems, and the environment. These pollutants may be released from both natural sources, such as wildfires, and human activities, including vehicle emissions, industrial facilities, power plants, construction, and fossil fuel combustion. Figure 6. highlights the different sources of air pollution. Common air pollutants include particulate matter (PM<sub>2.5</sub> and PM<sub>10</sub>), ground-level ozone, nitrogen dioxide, sulfur dioxide, and carbon monoxide. Fine particulate matter, especially PM<sub>2.5</sub>, is particularly dangerous because the particles are small enough to enter the lungs and bloodstream, increasing the risk of asthma, respiratory

---

<sup>13</sup> Firststreet.Org. "East Germantown, PA Flood Map and Climate Risk Report." Accessed May 9, 2026. [http://firststreet.org/neighborhood/east-germantown-pa/802870\\_fsid/flood](http://firststreet.org/neighborhood/east-germantown-pa/802870_fsid/flood).

infections, heart disease, stroke, and premature death. Ground-level ozone, often worsened during periods of extreme heat, can irritate the lungs and make breathing difficult, especially for children, older adults, and individuals with preexisting respiratory conditions. In urban communities such as Germantown, air pollution is often concentrated near major roads, industrial corridors, and densely developed areas, where residents may experience long-term exposure to poor air quality. Air pollution is also closely connected to environmental justice because low-income communities and communities of color are disproportionately exposed to higher pollution levels due to historic patterns of segregation, disinvestment, and zoning practices. As climate change intensifies heat waves and wildfire smoke events, improving air quality has become an increasingly important component of climate resilience and public health planning.

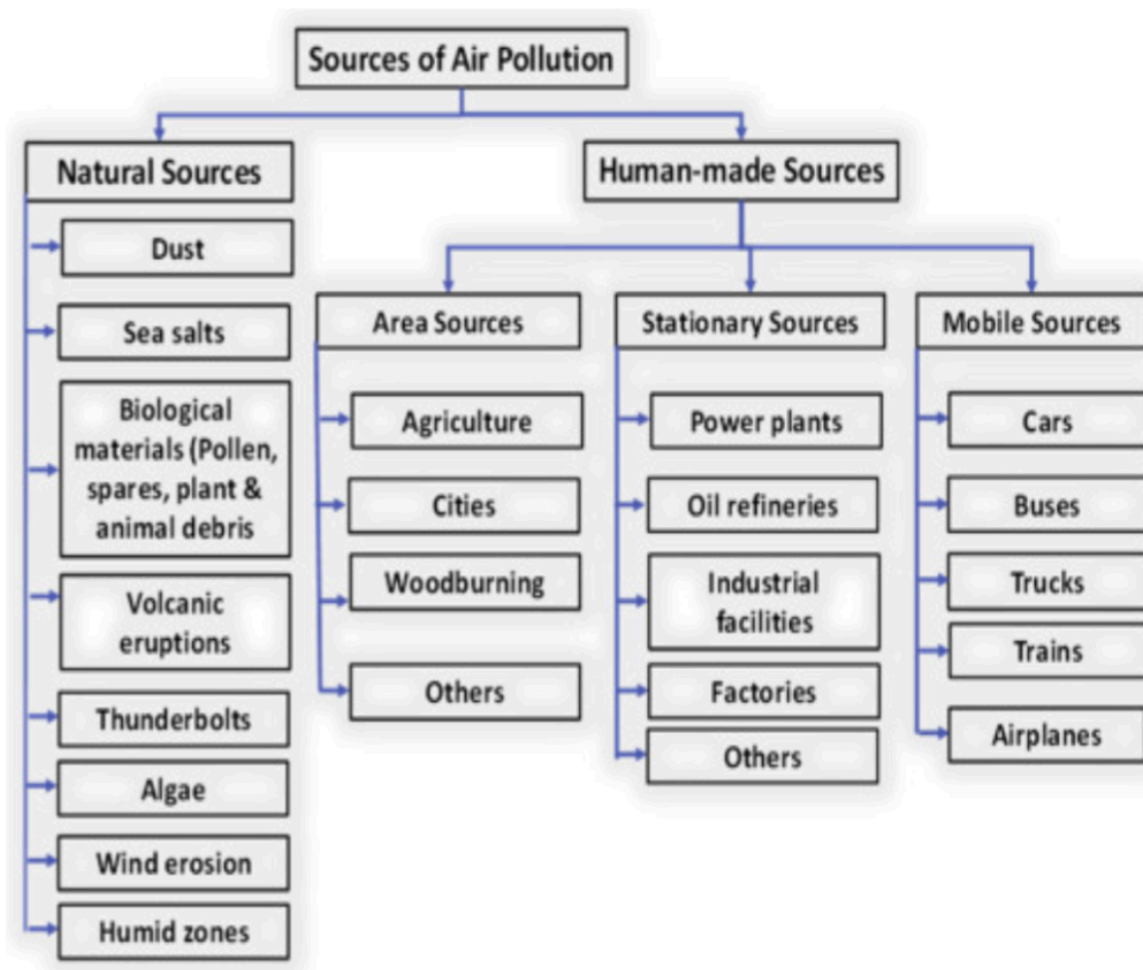


Figure 6. Sources of Air Pollution

Air pollution and air quality are essential considerations in climate resilience planning because they directly affect public health, environmental sustainability, and a community's ability to adapt to climate change. Poor air quality increases the risk of respiratory illnesses, cardiovascular disease, asthma, and premature death, particularly among vulnerable populations such as children, older adults, low-income residents, and individuals with preexisting health conditions. Climate change intensifies many forms of air pollution by increasing temperatures, worsening ground-level ozone formation, extending allergy seasons, and contributing to more frequent wildfires that release harmful smoke and fine particulate matter into the atmosphere. In

urban areas, these impacts are often compounded by traffic emissions, industrial pollution, and limited green space. Because extreme heat and air pollution frequently occur together, communities facing one climate-related hazard are often exposed to multiple overlapping health risks. For example, hotter temperatures can trap pollutants close to the ground, making breathing conditions more dangerous during heat waves.

Air quality is also closely tied to environmental justice, as historically marginalized communities are disproportionately located near highways, industrial facilities, and other pollution sources due to decades of discriminatory housing and zoning policies. Including air quality in climate resilience plans allows communities to develop strategies such as expanding tree canopy coverage, reducing transportation emissions, improving public transit, promoting clean energy, increasing air monitoring, and enhancing indoor air protection through filtration systems. These measures not only reduce pollution exposure but also strengthen overall community resilience, improve public health outcomes, and create healthier living environments in the face of climate change.

For Germantown, in particular, data from the Environmental Justice Screen (EJScreen) identified multiple overlapping sources of environmental pollution, including close proximity to hazardous waste facilities, high traffic density, diesel particulate emissions, and elevated ground-level ozone concentrations. These environmental burdens are illustrated in Figures 7 through 10 and demonstrate how residents are exposed to several forms of pollution simultaneously rather than in isolation. Among these factors, Germantown is especially impacted by vehicle-related emissions, particularly nitrogen dioxide (NO<sub>2</sub>) and fine particulate matter (PM<sub>2.5</sub>), both of which are commonly associated with traffic congestion, trucking routes, and

dense urban infrastructure. In addition, the neighborhood experiences regional ozone pollution, which is intensified during warmer months and prolonged heat events.

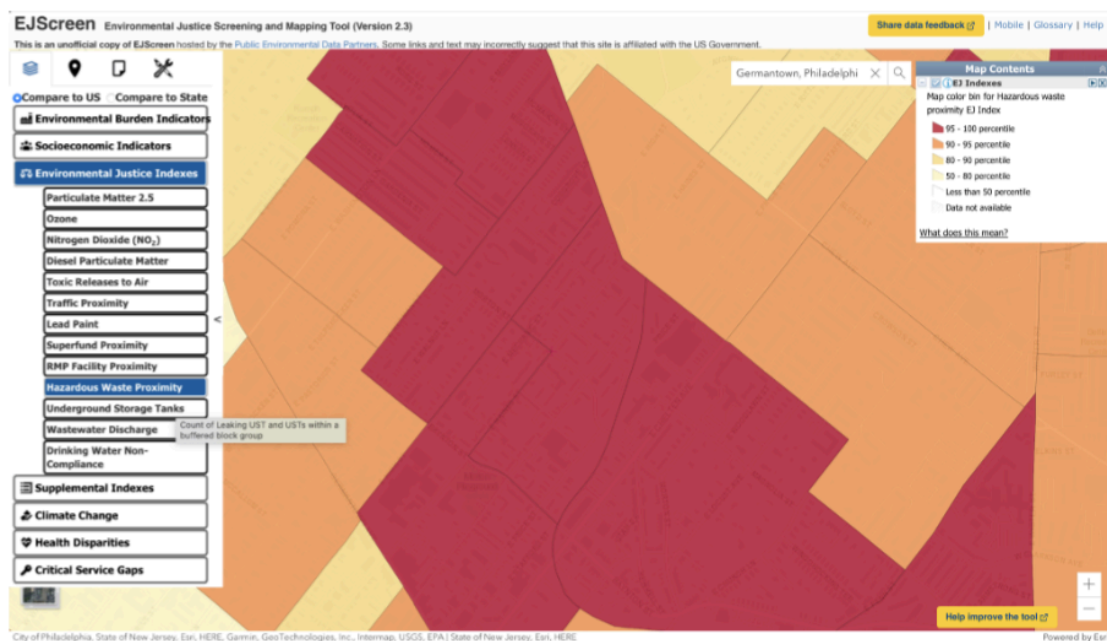


Figure 7. Hazardous Waste Proximity in Germantown, Pa by EJScreen

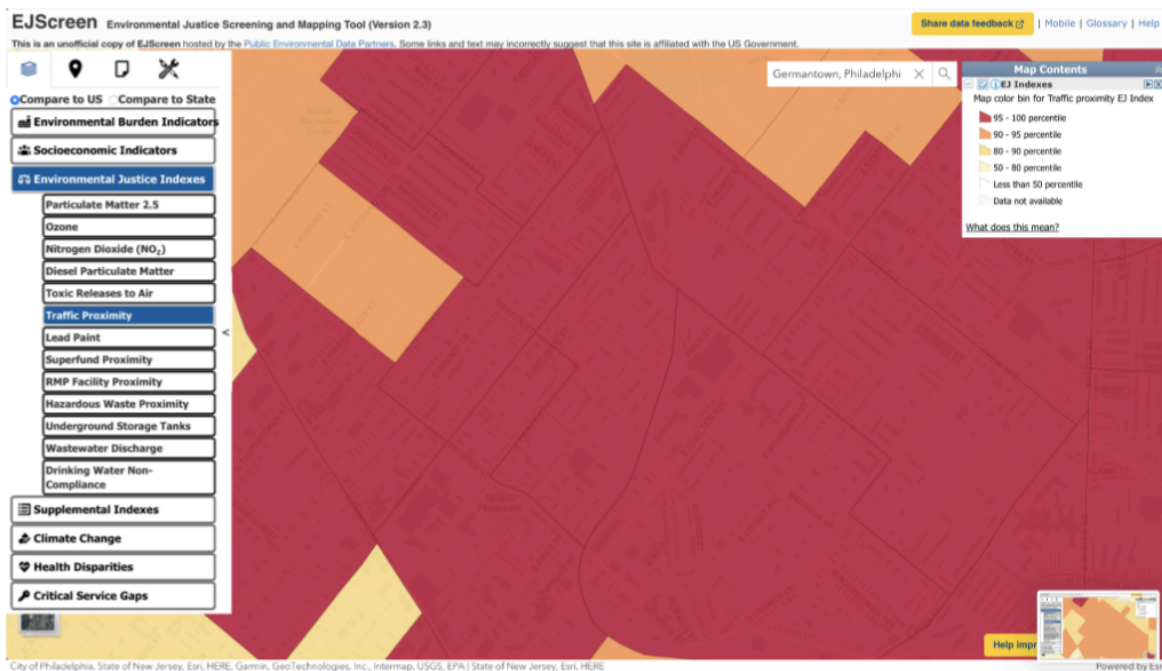


Figure 8. Traffic Proximity in Germantown, Pa by EJScreen

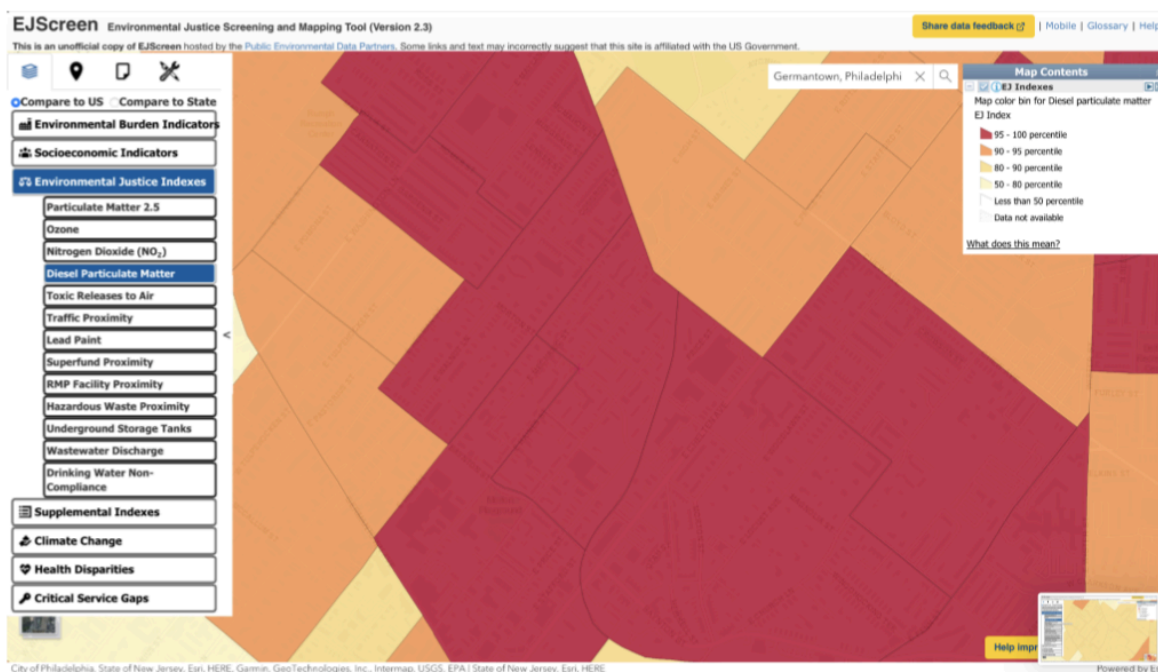


Figure 9. Diesel Particulate Matter in Germantown, Pa by EJScreen

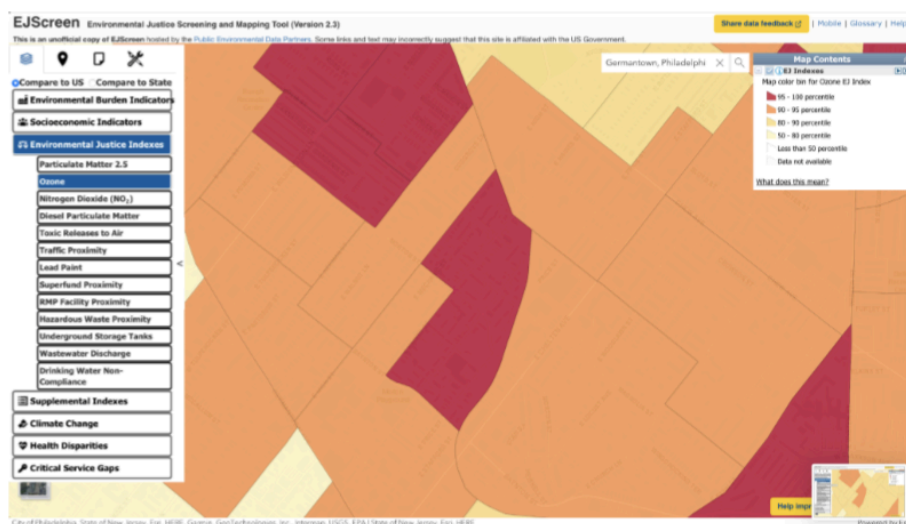


Figure 10. Ozone in Germantown, Pa by EJScreen

The significance of these findings becomes even more concerning when considered alongside Germantown's growing vulnerability to extreme heat. Heat and air pollution are deeply interconnected environmental hazards. Rising temperatures accelerate the chemical reactions that

form ground-level ozone, while stagnant hot air can trap pollutants closer to the ground, reducing air quality across urban neighborhoods. As a result, residents are often exposed to higher concentrations of harmful pollutants precisely during periods when their bodies are already under heat stress. This creates a compounding public health risk, particularly in urban heat islands where limited tree canopy, high levels of pavement, and aging infrastructure intensify local temperatures.

These environmental conditions disproportionately affect vulnerable populations within Germantown, including elderly residents, children, low-income households, and individuals with preexisting respiratory or cardiovascular conditions such as asthma, bronchitis, or heart disease. Children are especially susceptible because their lungs are still developing, while older adults are more likely to experience complications from prolonged heat exposure and poor air quality. For residents with asthma or other respiratory illnesses, exposure to PM<sub>2.5</sub>, NO<sub>2</sub>, and ozone can trigger breathing difficulties, asthma attacks, hospitalizations, and long-term declines in respiratory health. Furthermore, because Germantown is a historically marginalized and predominantly Black community, these environmental burdens reflect broader patterns of environmental injustice in which communities of color are more likely to face concentrated pollution exposure and fewer environmental protections.

### ***Heat***

Extreme heat refers to periods of unusually high temperatures that are significantly hotter than average conditions for a particular area and time of year. These events are often accompanied by high humidity, making it more difficult for the human body to cool itself and increasing the risk of heat-related illnesses such as heat exhaustion and heat stroke. Climate change has intensified extreme heat events by increasing global average temperatures and

making heat waves more frequent, longer-lasting, and more severe. In urban areas, the impacts of extreme heat are worsened by the urban heat island effect, a phenomenon in which cities become significantly hotter than surrounding suburban or rural areas. This occurs because materials commonly found in cities, such as asphalt, concrete, rooftops, and buildings, absorb and retain heat during the day and slowly release it at night. Limited tree canopy coverage and a lack of green space further reduce natural cooling. As a result, neighborhoods with fewer trees and more impervious surfaces can experience temperatures several degrees higher than nearby areas. The urban heat island effect disproportionately impacts low-income communities and communities of color, which have historically experienced disinvestment, fewer environmental resources, and higher exposure to pollution. Because of these combined risks, extreme heat has become a major public health and environmental justice concern in climate resilience planning.

Extreme heat is one of the most important issues addressed in climate resilience planning because it is the deadliest weather-related hazard in the United States and is becoming more frequent and severe as climate change intensifies. Urban neighborhoods such as Germantown are especially vulnerable due to the urban heat island effect, aging infrastructure, high population density, and limited tree canopy coverage. Prolonged exposure to extreme heat can increase rates of heat exhaustion, heat stroke, respiratory illness, cardiovascular disease, and death, particularly among vulnerable populations such as seniors, low-income residents, outdoor workers, and people with preexisting health conditions. Heat also worsens air pollution by increasing ground-level ozone formation, creating compounded environmental health risks for communities already experiencing social and economic inequities. Because of these growing dangers, climate resilience plans increasingly prioritize heat mitigation strategies such as tree canopy expansion, cooling centers, green infrastructure, and emergency response systems. At the federal level, the

Occupational Safety and Health Administration (OSHA) is currently moving toward what could become the first nationwide workplace heat safety standard through its proposed “Heat Injury and Illness Prevention in Outdoor and Indoor Work Settings” rule. The proposed regulation would require employers to implement heat safety plans, provide water and rest breaks, monitor heat conditions, and protect workers exposed to dangerous temperatures. Although the rule has not yet been finalized, its development reflects growing national recognition that extreme heat is both a public health emergency and a major climate resilience challenge requiring long-term policy intervention.

Germantown faces overlapping environmental and public health challenges caused by extreme heat, air pollution, aging infrastructure, and long-standing environmental inequities. These risks do not occur independently; rather, they interact and intensify one another, creating compounded impacts for residents. Philadelphia is experiencing increasingly frequent, longer, and more intense heat waves, with projections showing a dramatic rise in days above 100°F by the end of the century. Due to the urban heat island effect, temperatures can vary significantly across neighborhoods, with some areas measuring up to 22°F hotter than others. Dense built environments, limited green space, and heat-retaining materials such as asphalt and brick worsen local temperatures.

Germantown is particularly vulnerable because of its infrastructure makeup including high impervious surface coverage i.e. roads, asphalt, and rowhomes, uneven tree canopy distribution, older housing stock with poor insulation and limited cooling, and its proximity to traffic corridors and vehicle emissions.

## ***Food***

Modern food systems are increasingly brittle. Climate shocks, geopolitical instability, and economic pressures continue to unrelentingly disrupt global supply chains. Concomitantly, food systems across the world are presently dominated by large-scale, transnational actors that prioritize efficiency, consolidation, and minimizing costs over resilience. Events such as the Covid-19 pandemic and constant disruptions in global trade routes have revealed how quickly supply chains can unravel. When transportation corridors are blocked, labor forces disrupted, or processing facilities shut down, communities that depend on distant sources of food face immediate shortages and escalating prices. These disruptions are not anomalies but rather harbingers of a future in which climate change relentlessly intensifies systemic stress. Importantly, food systems are both a major contributor to climate change, accounting for roughly 37 percent of global greenhouse gas emissions, and one of the sectors most vulnerable to its impacts. Rising temperatures, extreme weather events, and ecological shifts threaten crop yields, food quality, and supply stability. As these dynamics escalate, the fragility of centralized, worldwide food systems becomes untenable.

In Germantown, these aforementioned global vulnerabilities manifest in hyper-localized ways. Indeed, the neighborhood is particularly exposed to climate-related disruptions that directly affect food access. Urban flooding is a persistent issue, compounded by buried waterways such as the Wingohocking Creek and an aging sewer system that increasingly struggles to absorb heavy rainfall. Flood events can block transportation routes, damage retail infrastructure, and compromise storage systems, profoundly interrupting the flow of food into and within the neighborhood. Power outages, often caused by storms and falling trees, pose an additional critical risk by disrupting refrigeration systems and further threatening food safety.

Heat waves, aggravated by the urban heat island effect, increase food spoilage and constrain the ability of residents to engage in outdoor labor, including urban agricultural practices, such as community gardens. These environmental pressures intersect with and magnify longstanding structural inequalities. While not classified as a food desert, Germantown is characterized by limited access to full-service grocery stores, with a retail landscape dominated by smaller corner stores offering fewer fresh and nutritious options. A limited number of grocery stores and markets means less availability of fresh, healthy, and nutritious food, and an increased consumption of nutritionally inadequate junk food. Food insecurity remains widespread, with a substantial portion of residents relying on programs such as the Supplemental Nutrition Assistance Program (SNAP), Women, Infants, & Children (WIC), and emergency food networks that can be and often are disrupted during crises.

### **Climate Resilience Strategies:**

#### *Introduction*

The following section outlines a series of proposed solutions and implementation strategies designed to strengthen climate resilience in Germantown while addressing the neighborhood's interconnected environmental, infrastructural, and social vulnerabilities. Because challenges such as flooding, extreme heat, air pollution, housing insecurity, and food system instability are deeply interconnected, no single intervention can fully address the risks facing the community. Instead, Germantown requires a comprehensive and multi-layered approach that combines green infrastructure, public health initiatives, housing improvements, equitable land-use planning, transportation reform, and community-based resilience efforts. These solutions are grounded in principles of environmental justice and hyperlocal planning,

recognizing that effective climate adaptation must respond directly to the specific historical conditions, environmental realities, and lived experiences of Germantown residents. Importantly, the proposed strategies are designed not only to reduce environmental risk, but also to improve public health, strengthen community capacity, expand economic opportunity, and support long-term neighborhood stability. Together, these recommendations form an integrated resilience framework that prioritizes equity, sustainability, and community participation throughout every stage of implementation.

### *Framework*

The implementation framework for Germantown's climate resilience plan should be organized into short-term, medium-term, and long-term phases in order to address immediate environmental and public health concerns while also building sustainable, long-lasting resilience. Because Germantown faces interconnected challenges related to extreme heat, air pollution, flooding, food insecurity, environmental injustice, and public health disparities, implementation must occur gradually and across multiple sectors at once. Dividing the plan into phases allows the neighborhood to respond to urgent risks immediately while also investing in structural and policy changes that will strengthen resilience over time.

In the short term, over the next one to three years, the priority should be protecting residents from immediate climate-related dangers while establishing the foundation for larger resilience efforts. Early actions should include expanding access to cooling centers in libraries, schools, recreation centers, and churches during extreme heat events. Public health outreach campaigns should also be developed to educate residents about heat safety, air quality alerts, asthma prevention, and available community resources. During this phase, Germantown can begin pilot tree-planting and urban greening projects in the hottest and least shaded areas of the

neighborhood, particularly near schools, bus stops, and heavily paved corridors. Additional short-term priorities include deploying community air-monitoring sensors, improving emergency heat communication systems through text alerts and neighborhood networks, distributing free or subsidized air conditioning units, and beginning small-scale housing weatherization projects for low-income households. These early interventions are important because they provide immediate relief to vulnerable residents while helping build community trust and participation.

The medium-term phase, covering approximately three to ten years, should focus on expanding infrastructure improvements and reducing long-term environmental exposure. During this period, Germantown should invest in large-scale building retrofits, including insulation upgrades, energy-efficient cooling systems, ventilation improvements, and indoor air filtration systems in homes, schools, and public housing. Green infrastructure projects should also expand significantly through increased tree canopy coverage, cool roofs, green roofs, permeable pavement, community gardens, and shaded pedestrian corridors. Transportation improvements are another major priority during this phase, including expanded public transportation access, safer bike infrastructure, anti-idling policies, and increased electrification of SEPTA buses and municipal vehicles to reduce traffic-related emissions. At the same time, community-based climate programs should continue growing through workforce development initiatives, climate health ambassador programs, and participatory planning efforts that involve residents directly in decision-making processes. These medium-term investments help reduce both environmental exposure and social vulnerability while strengthening neighborhood-wide adaptive capacity.

The long-term phase, extending beyond ten years, should focus on achieving transformational and permanent climate resilience within Germantown. Long-term goals include full restoration and equitable distribution of tree canopy coverage throughout the neighborhood,

climate-resilient redevelopment of housing and public infrastructure, and the permanent integration of environmental justice principles into zoning laws, land-use policies, and neighborhood planning processes. Future development projects should be required to include green space, heat-mitigating design, stormwater management systems, and protections against displacement or green gentrification. Over time, Germantown should aim to create a fully connected network of cooling spaces, green infrastructure, clean transportation systems, and resilient public facilities that can better withstand increasing climate pressures. Long-term monitoring of health outcomes, air quality, and neighborhood temperatures will also be necessary to evaluate progress and adapt strategies as climate conditions continue to change.

Successful implementation of this climate resilience plan will require collaboration among many different stakeholders, including the City of Philadelphia Office of Sustainability, community development corporations, local schools and universities, churches, healthcare providers, transit agencies, environmental organizations, and neighborhood residents themselves. Partnerships between government institutions and community groups are especially important because they help ensure that resilience strategies are both effective and responsive to local needs and lived experiences. Community participation is critical in identifying neighborhood priorities, building trust, and ensuring equitable distribution of resources and investments.

Although each component of the resilience plan, such as heat mitigation, air pollution reduction, flood resilience, transportation reform, and public health outreach, has its own specific implementation process, these strategies frequently overlap and reinforce one another. For example, expanding tree canopy coverage can simultaneously reduce temperatures, improve air quality, lower stormwater runoff, and improve mental and physical health outcomes. Similarly, weatherizing homes not only protects residents during heat waves but also reduces energy costs

and improves indoor air quality. Because these environmental and social challenges are deeply interconnected, Germantown's climate resilience plan must take an integrated and holistic approach that addresses multiple risks at the same time rather than treating each issue separately.

### *Metrics of Success*

The success of the climate resilience plan can be measured through a wide range of environmental, public health, infrastructure, and community-based indicators that reflect the interconnected challenges facing Germantown. These indicators should evaluate progress not only in reducing extreme heat and air pollution, but also in improving flood resilience, strengthening local food systems, increasing housing and infrastructure resilience, and advancing environmental equity.

Public health indicators may include reductions in heat-related illness and mortality, asthma attacks and respiratory hospitalizations, flooding-related health risks such as mold exposure, and other climate-related emergency room visits. Environmental indicators can include increased tree canopy coverage, lower neighborhood surface temperatures, improved air quality measurements such as PM<sub>2.5</sub>, NO<sub>2</sub>, and ozone levels, reduced stormwater runoff and localized flooding, and expanded green space throughout the neighborhood. Infrastructure and housing indicators may include the number of homes weatherized, the installation of cool roofs and green infrastructure, improved drainage systems, and increased access to cooling centers and clean indoor air spaces for vulnerable residents.

The plan should also measure progress in strengthening Germantown's food resilience and community capacity. This may include tracking the number of community gardens, urban farms, and fresh food access points established or expanded throughout the neighborhood, particularly in areas with limited access to healthy food. Additional indicators may include

participation in local climate programs, resident involvement in planning processes, workforce development opportunities in green infrastructure and environmental projects, and equitable distribution of climate investments across historically underserved areas.

Monitoring these outcomes over time is essential for understanding whether resilience strategies are effectively improving environmental conditions, reducing climate-related risks, and strengthening overall quality of life in Germantown. Long-term evaluation also allows policymakers, community organizations, and residents to adjust strategies as climate conditions evolve and neighborhood needs change.

## **Flooding**

### *Built Environment Solutions*

With the influence of climate change, rainfall patterns in Pennsylvania have shifted to result in more short-duration, high-intensity storms. These can more quickly overwhelm the drainage system and flood streets, transportation infrastructure, people's homes, and businesses, but the highly localized nature of the storms usually means that residents are left to deal with the damage using limited insurance reimbursements and their own resources, rather than the more extensive resources that would be accessible with a disaster declaration. This section will examine infrastructural techniques that reduce pluvial flooding, either by improving the drainage system or improving the rate at which water infiltrates the ground. The figure below showcases the various destinations of surface runoff, and strategies for alleviating the flow and volume of runoff as it goes into drainage systems.

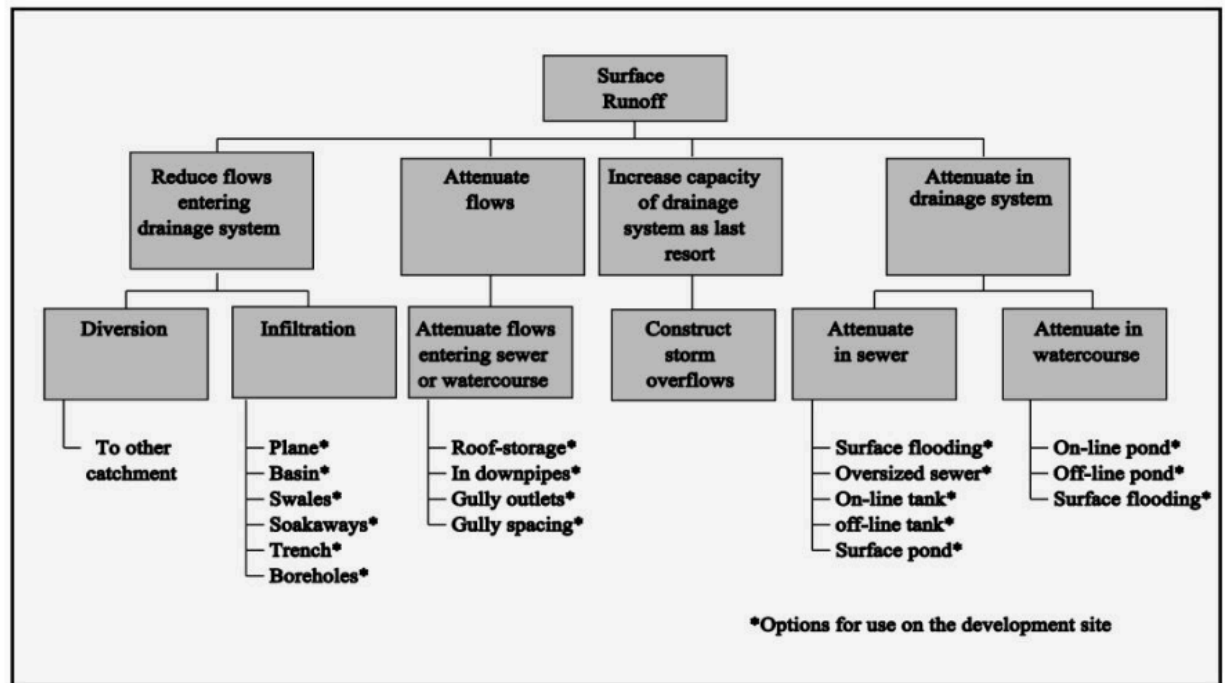


Figure 11: Surface Runoff Diagram

Generally strategies which reduce the amount of water going into the drainage systems are preferable to strategies which enlarge the amount that the drainage system can handle, because these strategies tend to have less capital expenditure and can help improve the sustainability of the overall system. These strategies can often focus around building a more Sustainable Urban Drainage System (SUDS), a strategy which aims to build an urban drainage system which increases permeability and can absorb water in a manner more similar to the way that natural systems do.

#### *Overflow/detention Basins*

A detention basin is a landscaped depression that is designed to manage stormwater runoff by temporarily storing water during heavy rain, and then slowly releasing it at a controlled rate to prevent downstream flooding and erosion. They can help mitigate both water quantity and quality impacts. This would help alleviate the threat of pluvial flooding from quick and intense

rainstorms that overwhelm the water table, by providing another place for the water to go before overwhelming the drainage systems. It would also provide water quality benefits by filtering out nutrients, heavy metals, and other pollutants as the water drains out through the basin.

Detention/overflow basins can also provide city-manicured greening if they are set up in spaces that were previously empty/uncared-for land. The largest complication with detention/overflow basins in Germantown is their need for undeveloped/unfilled space.

According to a GIS analysis of the Vacant Lot Indicator dataset from OpenDataPhilly<sup>14</sup> 65.5% of buildings in Germantown are within 50 meters of a currently vacant lot. This, of course, does not mean that a detention basin could be placed in every single one of these vacant lots. More in-depth analysis could be conducted to find what streets in Germantown, and even what specific drains overflow the most, and detention basins could be correspondingly built in the areas which have the most need and the best profile.

### *Infiltration trench*

The infiltration trench is another strategy that manages stormwater runoff by storing it so it can slowly percolate into the surrounding soil. Infiltration trenches are considered a Sustainable Drainage System strategy and can be used to reduce runoff volume and improve water quality. An infiltration trench is a gravel-filled, excavated trench which is placed at a point where water in a certain radius will drain towards it, so that they can provide temporary subsurface storage of stormwater runoff. Infiltration trenches can be much smaller than detention basins, and can be placed along the edge of any road or property that experiences flooding issues,

---

<sup>14</sup> OpenDataPhilly. "OpenDataPhilly." Accessed May 9, 2026. <https://opendataphilly.org/>.

but have a smaller radius of effect. They can be best utilized to alleviate issues and store stormwater runoff from impervious surfaces like roads, driveways, roofs, and parking.

Infiltration trenches reduce peak flow in sewer systems, and have a major benefit for water quality. Similar to detention basins, water that filters through the gravel or porous substrate in an infiltration trench leaves behind heavy pollutants, metals, etc.

However, also similarly to detention basins, in urban settings there can be issues of space for installation of infiltration trenches. For denser urban settings with tight sidewalk space between streets and buildings, infiltration trenches may not be as appropriate. These can be used by individual Germantown homeowners along the edge of their property to drain pluvial rain build-up before it threatens to overrun the drainage system, or can be implemented by the city along streets and parking lots wherever there is available land. Larger building owners, such as churches or business owners could also implement infiltration trenches next to parking lots and on land that runs adjacent to streets.

#### *Infiltration well*

Infiltration wells are another strategy to absorb runoff water before it reaches and overwhelms drainage systems. They take up less ground space than infiltration trenches, and can be ideal for small, confined spaces and deeper, concentrated flows such as collecting rooftop runoff.

#### National Programs/Resources:

*Federal Emergency Management Agency:* The Federal Emergency Management Agency (FEMA) is a United States Department of Homeland Security agency, created under President Jimmy Carter in 1978. The agency coordinates disaster responses across all levels of government

when there is a disaster event that overwhelms the resources of state and local authorities. For FEMA to offer aid, the governor of the state must declare a state of emergency and formally request that the federal government give aid through a Preliminary Damage Assessment showing that the disaster's severity exceeds state/local capacity. In order for the state to receive the aid, the President must then declare an emergency or major disaster, which enables different levels of FEMA assistance and unlocks aid for state/local/tribal governments and individuals. In 2018, the Disaster Recovery Reform Act was passed, which reformed FEMA's disaster programs to shift focus from reactive disaster response to proactive mitigation, helping communities become more resilient through emergency or disaster-related grant programs.<sup>15</sup>

*National Flood Insurance Program:* No section of Germantown is recognized as an official floodplain by FEMA, so Germantown residents are not required to purchase NFIP insurance, but it is a flood resource option available. The program has inherent environmental justice issues due to its prioritizing of property over human value, and also offers only reactionary support, not mitigatory. The National Flood Insurance Program (NFIP) provides flood insurance to property owners, renters, and businesses. It is managed by FEMA and delivered to the public through a network of more than 47 insurance companies. Municipalities must be members of the NFIP in order for their constituents to be eligible for flood insurance. Being a member of the NFIP has the requirement for municipalities to have a floodplain map and a Floodplain Management Ordinance to mitigate flooding effects. Municipalities can also join the Community Rating System, in which residents of municipalities receive discounts on flood insurance based on a rating of the municipality's flood resilience work, creating an

---

<sup>15</sup> Federal emergency management agency (FEMA)(b). (n.d.). The National Flood Insurance Program for Agents. Retrieved December 15, 2025, from <https://agents.floodsmart.gov/resource-library/en/policyholders-guide-severe-repetitive-loss>

incentivization structure for better flood mitigation and communication. NFIP currently has 4.7 million policyholders nationwide, and provides nearly \$1.3 trillion in flood coverage.<sup>16</sup>

*Building Resilience Infrastructure and Communities Grant:*

BRIC is a hazard mitigation program where states, U.S. territories, federally recognized Tribal Nations, and local governments can apply for federal funds for hazard mitigation activities. Unlike the NFIP, BRIC provides funds for pre-disaster and hazard mitigation projects, aiming to “save lives, protect property, and reduce the cost of future disaster response and recovery.”<sup>17</sup> Some potential activities can include upgrading and modernizing infrastructure, making it appropriate for larger-scale projects, such as the proposed project for daylighting the creeks. Because it can only be applied for by the city government, this is less accessible as a resource for Germantown residents and residential non-profits, but is still worth mentioning as a resource. The BRIC program is also in a precarious position – it was terminated by the Trump administration in April 2025, but was ordered to resume in March 2026 following legal challenges.

Local:

*Germantown Community Flood Risk Management Task Force*

In February 2020, the Philadelphia Water Department established the Germantown Community Flood Risk Management Task Force through the Flood Risk Management Task Force, with an aim to provide opportunities for Germantown community leaders to engage with

---

<sup>16</sup> *Flood insurance*. (b) (n.d.). FEMA.Gov. Retrieved December 15, 2025, from <https://www.fema.gov/flood-insurance>

<sup>17</sup> FEMA.Gov. “Building Resilient Infrastructure and Communities.” Accessed May 9, 2026. <https://www.fema.gov/grants/mitigation/learn/building-resilient-infrastructure-communities>.

City officials, and increase information sharing and flood preparation knowledge while reducing Germantown flood impacts. Meetings focused on covering the history of flooding in Germantown and distributing materials for residents, including visits from the Office of Emergency Management and the Department of License and Inspection. One focus of the task force was promoting flood insurance equitably in Germantown.

### *Green City, Clean Waters*

Green City, Clean Waters, is a 25-year plan by the City of Philadelphia to “reduce the volume of stormwater entering combined sewers using green infrastructure and to expand stormwater treatment capacity with traditional infrastructure improvements.”<sup>18</sup> It officially began in 2011 and is scheduled to complete in 2036 in accordance with their Long Term Control Plan. The program focuses on investing in new green stormwater projects, while also providing options for homeowners like free rain barrels and discounted residential landscaping upgrades to manage stormwater, like rain gardens. They also provide grants, credits, and other support to encourage sustainable stormwater management for new development and existing properties. In addition, they provide mini-grants, called the Soak It Up Adoption program, to organizations to help maintain neighborhood green tools. Organizations can use the grants to purchase materials and supplies, and hire people to help clean and monitor green stormwater infrastructure sites and engage with residents living nearby.

### *Nature-Based Solutions*

Nature-based solutions can be a valuable tool in environmental justice communities and areas facing disproportionate impacts from climate change. These projects are a cost-effective

---

<sup>18</sup> Sell, Lauren. “Green City, Clean Waters.” Philadelphia Water Department, April 9, 2021. <https://water.phila.gov/green-city/>.

strategy to strengthen climate resilience, while fostering shared economic and social prosperity within the community. As Germantown mainly faces issues of urban flash flooding and infrastructure flooding, often in homes, these solutions can help ease the burden from general impacts of climate change, such as urban heat island effect, along with promoting environmental and sustainability education throughout the community.

Conventional stormwater infrastructure (gutters, drains, pipes, tunnels, and retention basins) are designed to divert water to nearby bodies of water. As we know, Germantown infrastructure is not equipped to handle water and flooding, leading to damages to people's homes and buildings. Stormwater runoff is also an issue, picking up pollutants, ultimately increasing the pollution in nearby bodies of water. Nature-based solutions are essential to climate resilience strategies, building community and climate resilience through sustainable and environmentally friendly actions, fostering care for the surrounding environment and people.

### **Recommendations:**

Nature-based solutions and green infrastructure can limit the impacts of urban stormwater, flooding, addressing other challenges such as climate change, biodiversity loss, disaster risk, water and food security, human health, socioeconomic development, along with fostering community connection. In Germantown, there are a few examples of these nature-based solutions, specifically community gardens and composting initiatives. The city of Philadelphia has implemented a variety of solutions to combat stormwater pollution and the increasing impacts of climate change. This report provides recommendations of nature-based solutions Germantown to implement, along with a brief overview of the already existing green solutions.

### **Urban Reforestation:**

Urban reforestation is the process of utilizing trees and vegetation to mitigate the urban heat island effect, improve air quality, reduce stormwater runoff, increase biodiversity, and improve public health. It involves planting trees, creating green spaces, and integrating nature into urban planning. These trees not only provide an aesthetic value, but are also essential to combating climate change by acting as natural carbon sinks through sequestering carbon, and reducing the overall carbon footprint of urban areas<sup>19</sup>.

In 2020, the City of Philadelphia launched the ‘Philly Tree Plan’; a 10 year strategic plan for the planting and maintenance of Philadelphia’s tree canopy, guided by environmental justice, community engagement, and sustainability values<sup>20</sup>. This effort also involves new ways to work with residents in combating climate change, and supporting a more tree-friendly culture throughout the city.

#### Green Roof Installation:

Green roofs are a valuable tool to combat the impacts of climate change in an urban setting. It is the process of installing gardens/vegetation on rooftops to absorb rainwater, and increase greenery. This layer of vegetation over a flat or moderately pitched roof can provide runoff volume reduction and runoff peak rate attenuation<sup>21</sup>. Green roofs can also enhance building aesthetics and market value; regulate building temperatures; reduce urban heat island effect; improve air quality; educational benefits, and more. The Philadelphia Water Department lists green roofs in their Stormwater Management Guidance Manual, along with how to install and care for one.

---

<sup>19</sup>“evertreen.” 2025. Urban Reforestation: How Cities Can Combat Climate Change  
<https://www.evertreen.com/news/urban-reforestation-how-cities-can-combat-climate-change>.

<sup>20</sup> For more information on the plan: *Philly Tree Plan*. (n.d.). The Philadelphia City Fund.  
<https://philacityfund.org/programs/philly-tree-plan/>

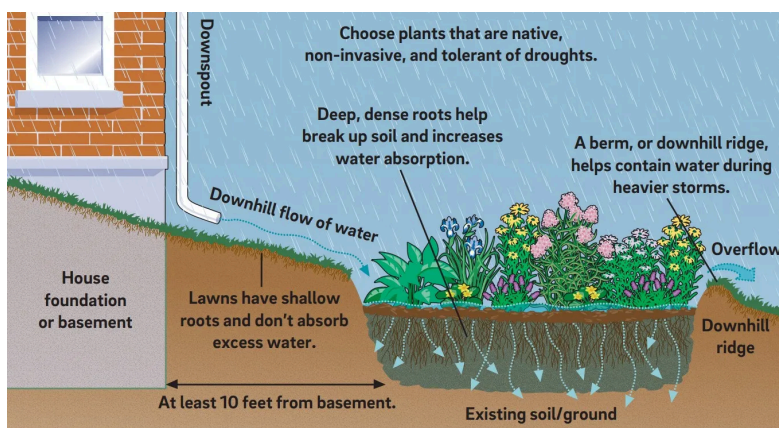
<sup>21</sup> *4.3 Green Roofs – Development Services*. (n.d.). Philadelphia Water Department. Retrieved April 25, 2026, from <https://water.phila.gov/development/stormwater-plan-review/manual/chapter-4/4-3-green-roofs/>



*example of a green roof in Philadelphia<sup>22</sup>*

### Rain Garden Installation:

Rain gardens are another valuable tool for stormwater management, collecting runoff from impervious surfaces, allowing water to directly enter the ground. Rain gardens generally include robust native plant species, thriving in extremely wet or dry weather. Rain gardens are typically lower than the surrounding ground level to collect rainwater from roads, driveways, and streets. These systems also effectively remove pollutants and reduce the amount of stormwater runoff<sup>23</sup>.



<sup>22</sup> 4.3 Green Roofs – Development Services. (n.d.). Philadelphia Water Department. Retrieved April 25, 2026, from <https://water.phila.gov/development/stormwater-plan-review/manual/chapter-4/4-3-green-roofs/>

<sup>23</sup> “Rain Gardens – Green Stormwater Infrastructure.” n.d. Philadelphia Water Department. Accessed May 8, 2026. <https://water.phila.gov/gsi/tools/rain-garden/>.

*Diagram of residential rain garden, courtesy of Philadelphia Water Department<sup>24</sup>*

### Community Gardens:

Community gardens are a form of green infrastructure, centering environmental justice and climate solutions through community engagement and greenery in urban areas. Community gardens reduce the effect of urban heat islands; provide ecosystem services; increase stormwater retention, all while supporting goals of trust-building, participation, and food security, which is especially important for marginalized groups. Community gardens not only provide food, but also enhance social empowerment, provide safe gathering spots, and promote restorative practices, which are all essential to communities facing disproportionate impacts of climate change. Urban community gardens can mitigate the impacts of food deserts and inaccessibility to healthy, fresh foods<sup>25</sup>.

Hansberry Garden & Nature Center is a community garden and hub for environmental education in Southwest Germantown. The center provides gardening and city harvest programs, along with various social and education events for the community. The garden provides a green space and the opportunity for community members to regularly maintain the garden, share knowledge, food, and company, and get more involved<sup>26</sup>.

### Stormwater Park Creation:

Stormwater parks are recreational spaces designed to flood during extreme weather events, and withstand flooding. These carefully created parks can reduce flooding elsewhere along with improving water quality. Stormwater parks mimic natural hydrology, with features

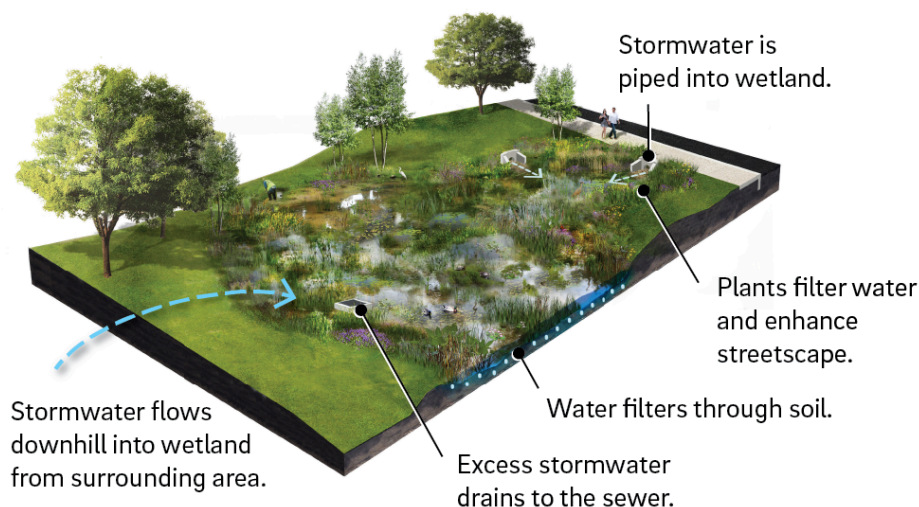
---

<sup>24</sup> “Rain Gardens – Green Stormwater Infrastructure.” n.d. Philadelphia Water Department. Accessed May 8, 2026. <https://water.phila.gov/gsi/tools/rain-garden/>.

<sup>25</sup> Scott, Heidi. 2022. “Sustainable Agriculture Lesson: Community Gardens – Justice, Safety, and Climate Solutions.” National Socio-Environmental Synthesis Center. <https://www.sesync.org/resources/sustainable-agriculture-community-gardens-justice-safety-and-climate-solutions>.

<sup>26</sup> n.d. Hansberry Garden & Nature Center – A community resource for fresh food and environmental education. Accessed May 8, 2026. <https://www.hansberrygarden.org/>.

such as constructed wetlands, bioswales, and rain gardens<sup>27</sup>. By utilizing green stormwater infrastructure rather than typical gray infrastructure, can cultivate healthier and more resilient



communities.

*Diagram of stormwater wetlands, courtesy of Philadelphia Water Department<sup>28</sup>*

### ***Extreme Heat and Air Pollution***

In the case of air pollution and extreme heat, several major goals were identified in order to address both the immediate and long-term environmental and public health challenges facing Germantown. These goals include reducing neighborhood temperatures and overall heat exposure, improving air quality and reducing residents' exposure to harmful pollutants, protecting vulnerable populations from climate-related health risks, improving housing quality and indoor climate resilience, increasing long-term adaptive capacity and community resilience, and advancing environmental justice through equitable investment and community leadership.

Reducing neighborhood temperatures is especially important because Germantown experiences the urban heat island effect, where dense infrastructure, asphalt, limited tree canopy,

<sup>27</sup> Frantz, George R. 2019. "Stormwater Management Park in Your Future?" American Society of Landscape Architects. <https://www.asla.org/news-insights/the-field/stormwater-management-park-in-your-future>.

<sup>28</sup> "Stormwater Wetland – Green Stormwater Infrastructure." n.d. Philadelphia Water Department. Accessed May 8, 2026. <https://water.phila.gov/gsi/tools/wetland/>.

and older buildings cause temperatures to rise significantly higher than in surrounding areas. Lowering heat exposure can help reduce heat-related illnesses, energy burdens, and stress on public health systems during extreme weather events. At the same time, improving air quality is critical because Germantown is heavily affected by traffic-related emissions, particulate matter, diesel pollution, and ozone exposure, all of which are worsened during periods of extreme heat.

Another central goal is protecting vulnerable populations, including elderly residents, children, low-income households, and individuals with preexisting respiratory or cardiovascular conditions. These groups face disproportionate risks during heat waves and poor air quality events due to social, economic, and health-related vulnerabilities. Improving housing and indoor climate resilience is therefore essential, particularly in Germantown's aging housing stock where many homes lack proper insulation, ventilation, cooling systems, or indoor air filtration.

The plan also emphasizes increasing long-term adaptive capacity and community resilience by strengthening local infrastructure, expanding green space, improving emergency response systems, and supporting community-based climate programs. Equally important is the goal of advancing environmental justice. Because Germantown is a historically marginalized and predominantly Black neighborhood, resilience strategies must ensure that investments, resources, and decision-making processes are equitable and community-driven. Residents should not only benefit from climate interventions but also play an active role in shaping them.

Using these interconnected goals as a framework, thirteen core solutions and strategies were developed to address the environmental, infrastructural, and social challenges facing Germantown. These solutions focus on areas such as urban greening, housing retrofits, air pollution reduction, public health outreach, cooling access, equitable land-use planning, and community engagement. Together, they form a comprehensive and integrated approach to

climate resilience that recognizes the relationship between environmental conditions, public health, infrastructure, and social equity. Table one highlights the thirteen proposed solutions and their corresponding actions, benefits, and implementation priorities.

| <b>Category</b>                                 | <b>Strategy</b>                 | <b>Key Actions</b>                                                                                                                                 | <b>Main Benefits</b>                                                                       | <b>Key Partners / Priority Groups</b>                                 |
|-------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Urban Cooling &amp; Green Infrastructure</b> | <b>1. Tree Canopy Expansion</b> | Plant trees in low-canopy corridors, heat-vulnerable blocks, schoolyards, vacant lots, and major streets; establish community stewardship programs | Reduces ambient temperatures; improves air quality; provides shade and stormwater benefits | Philly Tree Plan, community organizations, local schools and churches |
| <b>Urban Cooling &amp; Green Infrastructure</b> | <b>2. Cool Roofs, Green</b>     | Install reflective cool roofs on                                                                                                                   | Reduces indoor and outdoor                                                                 | City agencies, schools,                                               |

|                                                 |                                                             |                                                                                                            |                                                                            |                                                                  |
|-------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------|
|                                                 | <b>Roofs &amp; Cool Surfaces</b>                            | rowhomes;<br>develop green roofs on schools and public buildings; use permeable and reflective pavements   | temperatures;<br>lowers energy demand;<br>mitigates stormwater runoff      | housing organizations                                            |
| <b>Urban Cooling &amp; Green Infrastructure</b> | <b>3. Shade Infrastructure &amp; Vacant Land Activation</b> | Create tree-lined corridors, bus stop shade structures, pocket parks, and community gardens on vacant lots | Expands cooling access; improves walkability; increases green space equity | Community groups, transit agencies, neighborhood organizations   |
| <b>Housing &amp; Indoor Climate Resilience</b>  | <b>4. Weatherization &amp; Energy Retrofits</b>             | Expand insulation upgrades, air sealing, ventilation                                                       | Keeps homes cooler during heat waves; reduces                              | Seniors, low-income households, residents with chronic illnesses |

|                                                        |                                                             |                                                                                                                                                                                                                                                     |                                                                                                             |                                                                                           |
|--------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|                                                        |                                                             | improvements,<br>and<br>energy-efficient<br>cooling systems                                                                                                                                                                                         | utility costs and<br>energy burden                                                                          |                                                                                           |
| <b>Housing &amp;<br/>Indoor Climate<br/>Resilience</b> | <b>5.<br/>Cooling Access<br/>&amp; Emergency<br/>Relief</b> | Increase<br>cooling centers<br>in libraries,<br>schools, and<br>recreation<br>centers; extend<br>operating hours<br>during heat<br>emergencies;<br>provide<br>transportation<br>assistance,<br>utility<br>assistance, and<br>subsidized AC<br>units | Protects<br>vulnerable<br>residents during<br>extreme heat<br>events; improves<br>emergency<br>preparedness | Elderly<br>residents, medically<br>vulnerable<br>populations,<br>low-income<br>households |

|                                        |                                                                  |                                                                                                                                                                                                                                                                               |                                                                                                                                     |                                                                                                         |
|----------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Air<br/>Pollution<br/>Reduction</b> | <b>6. Traffic<br/>&amp; Emissions<br/>Reduction</b>              | <b>Expand</b><br><br>public transit,<br><br>bike and<br><br>pedestrian<br><br>infrastructure;<br><br>electrify SEPTA<br><br>buses and<br><br>municipal<br><br>fleets; enforce<br><br>anti-idling<br><br>policies; create<br><br>low-emission<br><br>zones near<br><br>schools | <b>Reduces</b><br><br>NO <sub>2</sub> and PM <sub>2.5</sub><br><br>exposure;<br><br>improves<br><br>neighborhood air<br><br>quality | <b>SEPTA,</b><br><br>City of<br><br>Philadelphia,<br><br>schools,<br><br>transportation<br><br>agencies |
| <b>Air<br/>Pollution<br/>Reduction</b> | <b>7.<br/>Community Air<br/>Monitoring<br/>(Citizen Science)</b> | <b>Deploy</b><br><br>neighborhood-<br><br>level air quality<br><br>sensors; partner<br><br>with residents,<br><br>schools, and<br><br>community<br><br>organizations                                                                                                          | <b>Fills gaps</b><br><br>in official<br><br>monitoring;<br><br>provides<br><br>hyperlocal<br><br>pollution data;<br><br>strengthens | <b>Residents,</b><br><br>schools, community<br><br>organizations                                        |

|                                              |                                                        |                                                                                                                              |                                                                                        |                                                                              |
|----------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
|                                              |                                                        |                                                                                                                              | community engagement                                                                   |                                                                              |
| <b>Air Pollution Reduction</b>               | <b>8. Indoor Air Protection</b>                        | Promote HEPA filtration in homes, schools, and public housing; provide education during high ozone and poor air quality days | Reduces indoor pollutant exposure; protects vulnerable residents during extreme events | Schools, public housing agencies, healthcare organizations                   |
| <b>Public Health &amp; Social Resilience</b> | <b>9. Heat &amp; Air Quality Early Warning Systems</b> | Integrate heat alerts and AQI alerts; expand outreach through text alerts, faith-based networks, and                         | Improves emergency communication and preparedness; protects vulnerable populations     | Elderly residents, medically vulnerable populations, community organizations |

|                                              |                                                    |                                                                                                                                                              |                                                                                         |                                                      |
|----------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------|
|                                              |                                                    | community organizations                                                                                                                                      |                                                                                         |                                                      |
| <b>Public Health &amp; Social Resilience</b> | <b>10. Community Health Outreach</b>               | Train<br>“climate health ambassadors”<br>to check on residents,<br>distribute water and educational materials, and<br>conduct outreach during extreme events | Builds<br>community resilience;<br>improves public health awareness and support systems | Local<br>residents, healthcare providers, nonprofits |
| <b>Land Use, Planning &amp; Equity</b>       | <b>11. Heat-Resilient Zoning &amp; Development</b> | Require<br>tree canopy minimums,<br>green space requirements, and<br>heat-mitigating building design;<br>encourage                                           | Creates<br>long-term neighborhood resilience;<br>reduces future heat exposure           | City<br>planners, developers, environmental agencies |

|                                            |                                                                   |                                                                                                                             |                                                                              |                                                                  |
|--------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------|
|                                            |                                                                   | climate-resilient<br>redevelopment                                                                                          |                                                                              |                                                                  |
| <b>Land Use,<br/>Planning &amp; Equity</b> | <b>12.<br/>Anti-Displacement &amp; Equity<br/>Protections</b>     | Implement property tax relief, affordable housing protections, and equitable investment strategies                          | Prevents displacement and green gentrification; supports long-term residents | Long-term residents, housing organizations, local government     |
| <b>Land Use,<br/>Planning &amp; Equity</b> | <b>13.<br/>Participatory Planning &amp; Workforce Development</b> | Involve residents in planning decisions; create local jobs in urban forestry, green infrastructure maintenance, and climate | Strengthen community leadership; supports local economic development         | Residents, workforce organizations, community development groups |

|  |  |                        |  |  |
|--|--|------------------------|--|--|
|  |  | resilience<br>programs |  |  |
|--|--|------------------------|--|--|

Table One. A Table of Strategies, Actions, Benefits, and Groups for Extreme Heat and Air Pollution

*Strategies and Actions*

Addressing extreme heat and air pollution in Germantown requires a comprehensive, multi-layered approach that combines improvements to the built environment, public health interventions, land-use planning, and community engagement. Because Germantown experiences both environmental and social vulnerability, solutions must prioritize equity and focus on residents who are most at risk, including low-income households, seniors, children, and people with preexisting health conditions.

One of the most important built environment interventions is urban greening and tree canopy expansion. Increasing tree coverage is one of the most effective ways to reduce urban heat island effects because trees provide shade, cool surrounding air through evapotranspiration, and improve overall air quality. Priority should be given to heat-vulnerable blocks with little existing canopy, particularly around schoolyards, major streets, and vacant lots. These areas often experience the highest surface temperatures during summer months. This strategy would fall primarily under the medium-term and long-term implementation phases. In the short term, the city could launch pilot tree-planting projects in partnership with local nonprofits and neighborhood organizations. Over the long term, Germantown could work toward full canopy restoration through sustained investment and community stewardship programs that train residents to maintain and protect newly planted trees.

Another major strategy involves cool surfaces and green infrastructure improvements. Installing cool roofs with reflective coatings, green roofs, and permeable pavement can significantly reduce surface temperatures and lower indoor heat exposure for residents. Green

infrastructure also helps absorb stormwater runoff, creating additional environmental benefits for neighborhoods already vulnerable to flooding and aging infrastructure. These projects would largely fall under the medium-term implementation framework because they require planning, funding, and coordination between property owners, city agencies, and developers. Over time, integrating these technologies into both public and private buildings could substantially improve neighborhood climate resilience.

Building retrofits are also essential for protecting residents from dangerous indoor heat exposure. Many homes in Germantown are older and lack proper insulation, weatherization, or energy-efficient cooling systems. Retrofitting homes with better insulation, updated windows, and efficient air conditioning systems can lower utility costs while protecting residents during extreme heat events. Priority should be given to seniors and low-income households, who are often least able to afford cooling costs. These upgrades would primarily occur during the medium-term phase, although some emergency weatherization assistance could begin in the short term through existing federal and local programs.

Public health and emergency response systems are another critical component of climate resilience planning. Philadelphia already operates a Heat Health Emergency Program, but Germantown would benefit from expanded outreach and communication efforts. A stronger heat early warning system could distribute alerts through text messages, community organizations, churches, and local faith-based networks to ensure residents receive timely information during dangerous heat events. Because these measures can be implemented relatively quickly, they would fall within the short-term implementation period.

Increasing access to cooling resources is equally important. Cooling centers should be expanded throughout Germantown, with extended operating hours at libraries, recreation centers,

and public buildings during heat emergencies. Transportation assistance should also be provided so that elderly or disabled residents can safely access cooling sites. Utility assistance programs are another important short-term strategy because many households avoid using air conditioning due to high electricity costs. These measures provide immediate protection against heat-related illness and mortality while longer-term infrastructure projects are developed.

Community health outreach can further strengthen neighborhood resilience. Training local “heat ambassadors” to check on vulnerable residents, distribute water, and share public health information can help reduce isolation during heat emergencies. Community-based outreach is especially effective because residents often trust local leaders and neighbors more than outside institutions. This initiative would begin in the short term and continue throughout all implementation phases as a long-term community resilience strategy.

Improving air quality is another major goal of the climate resilience plan. Transportation-related pollution contributes significantly to poor air quality in Philadelphia, particularly near major roads and traffic corridors. Expanding public transportation access, improving bike infrastructure, promoting electric vehicles, and enforcing anti-idling regulations can help reduce emissions over time. These actions would primarily occur during the medium-term implementation phase because they require infrastructure investment and policy coordination. However, anti-idling enforcement and public education campaigns could begin immediately.

Local air quality monitoring would strengthen public awareness and environmental accountability. Installing neighborhood-level air quality sensors throughout Germantown would allow residents to access real-time data on pollutants such as PM<sub>2.5</sub> and ozone. This information could help residents make informed decisions during high pollution days while also identifying

environmental hotspots that require additional intervention. These monitoring systems could begin in the short term and expand in the medium term as more funding becomes available.

Indoor air protection is also essential, especially because residents spend much of their time indoors during extreme weather events. Schools, homes, and community centers should promote the use of HEPA filtration systems and distribute educational materials about reducing indoor pollution exposure during high ozone or wildfire smoke events. These interventions would begin in the short term through public health campaigns and continue into the medium term as filtration systems become more widely available.

Land-use and zoning policies can also shape Germantown's long-term climate resilience. Heat-resilient zoning policies should require minimum tree canopy coverage and green space in new developments to prevent future environmental inequities. These policy changes would largely occur during the long-term implementation phase because zoning reform requires legislative action and long-term planning. However, planning discussions and policy development could begin immediately.

Vacant land activation presents another opportunity for environmental and community improvement. Many vacant lots in Germantown could be transformed into pocket parks, gardens, or green community spaces that provide shade, reduce heat absorption, and improve neighborhood aesthetics. These projects could begin in the short term through pilot programs and expand over time into a broader long-term redevelopment strategy.

Community engagement and equity must remain central throughout every stage of implementation. Participatory planning processes should directly involve residents in decisions about tree planting locations, cooling center placement, and neighborhood priorities. Residents are most familiar with the environmental challenges affecting their communities, making their

input essential for successful planning. Community participation should begin immediately and continue throughout all phases of implementation.

Workforce development programs also provide an opportunity to connect climate resilience with economic justice. Creating local jobs in urban forestry, green infrastructure maintenance, weatherization, and environmental monitoring can ensure that Germantown residents directly benefit from climate investments. These workforce initiatives would primarily develop during the medium-term implementation phase while continuing long term as climate adaptation projects expand.

### *Key Takeaways*

Notably, understanding the intersection of extreme heat and air pollution is therefore essential when developing a hyperlocal climate resilience plan for Germantown. Addressing only one environmental issue without considering the other would fail to capture the cumulative nature of the risks residents face on a daily basis. Strategies such as increasing tree canopy coverage, reducing traffic emissions, expanding green infrastructure, improving public transit, and creating accessible cooling centers can help mitigate both heat and air pollution simultaneously. By focusing on these interconnected challenges at the neighborhood level, Germantown can develop more equitable and effective resilience strategies that directly respond to the lived environmental realities of its residents.

### *Funding*

Implementing the proposed climate resilience strategies in Germantown will require substantial financial investment and long-term institutional support, making federal funding opportunities an essential component of the plan's success. Several federal agencies already

administer programs that align closely with the environmental, infrastructural, and public health needs identified throughout this project.

The Environmental Protection Agency (EPA) offers Environmental Justice Grants, which could support community air monitoring initiatives, citizen science projects, public health outreach, and environmental health equity programs aimed at reducing pollution exposure in Germantown. Additionally, the EPA's Climate Pollution Reduction Grants (CPRG) program could assist with local climate planning, emissions reduction strategies, and neighborhood-scale sustainability initiatives. The Federal Emergency Management Agency (FEMA) also provides major resilience funding through the Building Resilient Infrastructure and Communities (BRIC) program, which could support green infrastructure improvements, urban cooling investments, stormwater management systems, heat mitigation projects, and other climate adaptation measures designed to reduce long-term disaster vulnerability.

Housing-related resilience interventions may qualify for support through the U.S. Department of Housing and Urban Development (HUD), particularly through Community Development Block Grants (CDBG), which can fund housing retrofits, cooling centers, accessibility improvements, and broader neighborhood revitalization projects. HUD's HOME Investment Partnerships Program could further support the development and preservation of affordable climate-resilient housing for low-income residents.

Energy-related resilience strategies could receive funding from the Department of Energy (DOE) through programs such as the Weatherization Assistance Program (WAP), which helps improve energy efficiency in low-income households, and the Energy Efficiency and Conservation Block Grant (EECBG) program, which supports local clean energy and energy conservation initiatives.

Transportation and mobility improvements, including pedestrian safety infrastructure, public transit access, and reconnecting historically divided neighborhoods, may qualify for funding through the Department of Transportation (DOT), including the Reconnecting Communities Pilot Program and Safe Streets for All initiative.

Finally, urban greening efforts such as tree canopy expansion, reforestation, and community forestry projects could receive support through the USDA Urban and Community Forestry Program. Together, these funding opportunities demonstrate that many of the resilience strategies proposed for Germantown are not only theoretically achievable, but also potentially financially attainable through coordinated federal, local, and community partnerships.

### ***Food Systems***

Overall, building a climate-resilient food system in Germantown, requires a structural rethinking of how food is produced, distributed, accessed, and governed at the neighborhood level. In an era increasingly marked by climate volatility, supply chain vulnerability, and ever-deepening inequalities, Germantown offers both a warning and an opportunity. Its vulnerabilities, which are rooted in disinvestment, environmental exposure, and reliance on external food systems, are both real and urgent. However, its strengths, especially the dense networks of mutual aid, community leadership, and emerging local food infrastructure, provide a powerful foundation for progress and transformation. Building a climate-resilient food system in this community is not merely about ensuring food access during crises, but also about constructing a durable, equitable, and local system capable of withstanding and adapting to an uncertain future.

It must be noted Germantown is not starting from scratch. The neighborhood presently contains a rich ecosystem of community organizations, grassroots initiatives, and emerging food

enterprises that can serve as anchors for resilience. Organizations such as Germantown Residents for Economic Alternatives Together (GREAT) are already working to establish resilience hubs and strengthen mutual aid networks. Food Moxie has also developed powerful programs in urban agriculture education and food production, teaching residents the skills and resources needed to participate in the local production of their food. Simultaneously, the expansion of cooperative retail models such as Weavers Way Co-op has markedly improved access to locally sourced and organic foods while maintaining economic value within the community. Notably, before the Weavers Way Co-op, as much as 80% of Germantown residents shopped for their food in stores outside of the neighborhood, leading to a striking food dollar leakage in the community. With Weavers Way increasing access to fresh foods, this leakage has decreased, although an exact measure of this impact remains to be seen. These efforts, combined with active environmental programming and community engagement, create a strong foundation for a more resilient and equitable food system.

The vision for Germantown's food future is a distributed, community-controlled system that maintains access during climate shocks, expands local production and storage capacity, builds economic and cultural ownership, and reduces dependence on fragile consolidated supply chains. Achieving such a vision requires coordinated action across multiple strategic pillars outlined below, each addressing a distinct dimension of fragility, while concomitantly reinforcing each other.

The first pillar centers on food distribution, access, and emergency preparedness. In a climate-disrupted future, the ability to maintain food access during shocks is paramount. Thus, establishing food resilience hubs at trusted community institutions, such as schools, churches, and recreation centers, would provide critical infrastructure during emergencies. These hubs

would contain solar-powered backup energy systems, battery storage, all of which powers the infrastructure's refrigeration capacity to ensure that perishable foods remain safe during outages. Community fridges and cold storage facilities would serve both everyday needs and emergency situations. At the same time, such locations would also maintain stockpiles of shelf-stable foods which would provide several days' worth of sustenance during disruptions and serve residents on a daily basis. Importantly, these hubs would not operate in isolation but as part of a broader network of mutual aid, coordinated through organizations like GREAT. Training residents in household-level preparedness, such as safe food storage, water access, and emergency planning, would further decentralize resilience, ensuring that individual households are capable of navigating disruptions. Organizations such as the Watershed and GREAT could be charged with educational programs to tackle that aspect of this pillar.

The second pillar focuses on climate-resilient urban agriculture. Expanding local food production is an essential non-negotiable criteria for reducing dependencies on external supply chains and increasing access to fresh, nutritious foods. Germantown's vacant lots and other identified underutilized spaces present powerful opportunities for the development of micro-farms and community gardens. However, such systems must be designed with climate adaptation in mind. For instance, raised plotting beds can mitigate the impacts of flooding, protecting crops, while shade structures and drought-resistant crops can address urban heat stress. Rainwater harvesting systems can reduce reliance on municipal water supplies, further enhancing sustainability and resilience. Programs modeled after those developed by Food Moxie can provide training and employment opportunities, particularly for youth, creating pathways into the green economy. By prioritizing culturally relevant crops and diverse growing practices,

urban agriculture can also strengthen community identity and dietary diversity, both of which are critical components of resilience.

The third pillar addresses the resilience of food retail and supply chains. As previously mentioned, Germantown's retail landscape is currently characterized by a lack of full-service grocery stores and a reliance on smaller stores that may also lack the infrastructure needed to withstand disruptions. Supporting cooperative and community-owned grocery models can help address these gaps while ensuring that economic benefits remain within the neighborhood.

Investments in existing small businesses, such as upgrading refrigeration systems and installing backup power systems, can enhance their capacity to operate during emergencies. At the same time, the development of local aggregation and distribution hubs can connect urban farms and regional producers with local retailers and institutions that buy and sell food items. These hubs, supported by investments, can reduce bottlenecks in the supply chain and create more flexible, adaptive systems capable of responding to disruptions.

The fourth pillar of this food resilience section within this climate resilience plan centers around infrastructure and land use, highlighting the profound linkages between environmental systems and food systems. Expanding green stormwater infrastructure, such as rain gardens, bioswales, and permeable surfaces, can reduce flooding while creating more hospitable conditions for urban agriculture. Concomitantly, protecting land for food production through zoning policies ensures that these gains are not lost to competing development pressures. Moreover, integrating food systems planning into broader watershed management efforts can further align environmental and food resilience goals, particularly in relation to the Wingohocking Creek and other local waterways.

The fifth pillar focuses on economic and social resilience. Food insecurity is fundamentally tied to income and access, not just supply. Supporting food-based livelihoods, such as urban farming, food processing, and cooperative businesses can create stable employment opportunities while strengthening the local economy. Expanding access to programs such as the Supplemental Nutrition Assistance Program, and pairing these benefits with local matching initiatives can increase purchasing power while supporting local producers. Community purchasing cooperatives can further reduce costs and improve access to healthy foods, while fostering collective ownership and decision-making.

Importantly, implementation of this vision requires a phased approach. In the short term, efforts should focus on piloting resilience hubs, mapping food system assets and vulnerabilities, and establishing emergency food reserves. In the medium term, scaling urban agriculture networks, retrofitting food retailers, and developing aggregation and distribution infrastructure will be critical. In the long term, institutionalizing neighborhood-level governance structures and integrating Germantown's food system into broader citywide resilience planning will ensure sustainability and adaptability.

A critical component of this transformation is the strategic leveraging of funding sources and policy frameworks. Federal programs administered by the United States Department of Agriculture play a central role in supporting local food system development. This climate resilience plan highlights below several of the most promising funding programs, grants, and assistance resources that may be leveraged to carry out this vision.

More broadly, Philadelphia and the larger Pennsylvania region provide a strong example of how targeted policies and programs can support the development of resilient local food systems. At the state level, recent investments reflect a growing recognition that agriculture and

food infrastructure are central to economic development and climate adaptation. Governor Josh Shapiro's economic strategy prioritizes agriculture and manufacturing as key drivers of the state's future. Significant funding has been directed toward programs that enhance local supply chains, including the Agriculture Innovation Grant Program, the Pennsylvania Agricultural Surplus System, and the State Food Purchase Program. These initiatives collectively support farmers, expand access to markets, and address food insecurity, particularly for low-income populations.

One of the most impactful federal-state partnerships is the Resilient Food Systems Infrastructure (RFSI) Program, administered by the U.S. Department of Agriculture (USDA). Launched in 2023, RFSI focuses on strengthening the "middle" of the supply chain, the often-overlooked stages between production and consumption, such as processing, aggregation, and distribution. By funding infrastructure and equipment, the program enables small and mid-sized producers to access new markets and create value-added products.

In Pennsylvania, RFSI has already delivered over \$21 million in grants to farms and food businesses. A notable example in Philadelphia is a project led by The Food Trust, which received funding to expand cold storage, commercial kitchen space, and transportation capacity. This investment allows local aggregators to bring over 100 additional products from dozens of regional producers into Philadelphia markets. Such infrastructure improvements are critical for scaling local food systems, as they reduce bottlenecks and enable small producers to compete more effectively.

Another key initiative is the Fresh Food Financing Initiative (FFFI), which addresses both supply chain development and food access. By investing in grocery stores, cooperatives, and food distribution businesses, FFFI strengthens the connection between local producers and

underserved communities. The program supports essential infrastructure such as refrigeration, transportation, and food safety systems, all of which are necessary for maintaining a reliable and equitable food supply.

FFFI also plays an important role in addressing food insecurity. In Pennsylvania, more than one in nine residents experiences food insecurity, often compounded by limited access to fresh and nutritious foods. By bringing locally sourced products into neighborhoods that have historically lacked grocery stores or fresh food options, the initiative helps close this gap while supporting regional farmers. This dual focus on economic opportunity for producers and improved access for consumers makes FFFI a cornerstone of resilient food system development.

At the city level, the Philadelphia Food Justice Initiative (PFJI) complements these efforts by focusing on community-driven solutions. Administered by the city's Department of Public Health in partnership with The Reinvestment Fund, PFJI provides grants to local organizations working to advance food justice. Since its launch, the initiative has distributed millions of dollars to support projects that increase access to culturally relevant, affordable, and nutritious food. PFJI is particularly notable for its emphasis on equity. It prioritizes projects led by communities that have historically experienced health and economic disparities. By supporting grassroots organizations, the initiative ensures that solutions are tailored to the specific needs of different neighborhoods. This localized, participatory approach strengthens not only the food system but also community cohesion and empowerment.

Together, these programs illustrate a comprehensive strategy for building resilient local food systems in Philadelphia. Investments in infrastructure, funding for innovation, and support for community-led initiatives all contribute to a more distributed and adaptable system. As climate change and global instability continue to pose challenges, Philadelphia's approach offers

a model for other cities. While Germantown currently has food system vulnerabilities, the community also has notable potential and opportunity for mitigating such risks and developing localized, diversified food supply chains and distribution networks, simultaneously improving food and community resilience. By prioritizing local capacity, fostering collaboration between public and private sectors, and centering community needs, the community is taking meaningful steps toward a food system that is not only more resilient but also more just and sustainable.

**Community Engagement Strategy:**

The next step in this climate resilience plan is how to engage with the community, share the information we learned, and get people more involved in increasing and maintaining community resilience. Other capstone groups designed curriculum for water quality testing and flooding for The WaterShed. These educational tools are specifically important for educating younger generations, increasing community knowledge and hands-on research into local environmental issues.

Other community engagement strategies include increased citizen science, such as through the Senior Environment Corps, or through the water quality curriculum, which allows for students to test the water quality throughout Germantown. Workshops are also an important strategy for community engagement, to get people involved and help them learn about issues facing their community, and what the possible solutions are. Lastly, we created preliminary pamphlets for The WaterShed and Germantown community members on rain gardens, the fundamentals of air pollution and its community-based solutions, and the most promising USDA grants for food resilience initiatives.

**Conclusion:**

Climate change presents an urgent and uneven challenge for neighborhoods such as Germantown, where environmental risks are deeply intertwined with historical inequities, aging infrastructure, public health disparities, and economic vulnerability. Flooding, extreme heat, air pollution, food insecurity, and energy burden do not occur independently; instead, they operate as interconnected systems that compound one another and disproportionately affect historically marginalized communities. Germantown's vulnerability to climate change is therefore not simply the result of environmental exposure alone, but also the consequence of decades of structural racism, disinvestment, discriminatory housing policies, and uneven access to environmental resources.

This project demonstrates the importance of hyperlocal climate resilience planning as an alternative to broad, one-size-fits-all approaches to climate adaptation. While citywide resilience strategies remain important, they often fail to capture the unique environmental conditions, social vulnerabilities, and lived experiences present within individual neighborhoods. Germantown illustrates why neighborhood-level analysis is essential for identifying concentrated climate risks, understanding compounding vulnerabilities, and developing targeted solutions that reflect local realities. Hyperlocal planning also recognizes the importance of place-based knowledge and community participation, acknowledging that residents themselves possess critical insight into how environmental risks shape daily life.

Throughout this project, environmental justice remained a central framework for understanding both vulnerability and resilience. Climate change does not affect all residents equally, and resilience planning must therefore prioritize those facing the greatest environmental and social burdens. Effective climate adaptation requires more than physical infrastructure

improvements; it also demands investments in public health, affordable housing, community networks, accessible information, and equitable decision-making processes. Without intentional protections, even well-intentioned sustainability initiatives can contribute to displacement and green gentrification, reproducing the very inequalities they seek to address.

The resilience strategies proposed in this plan emphasize a comprehensive and integrated approach to adaptation. Urban greening, tree canopy expansion, weatherization programs, cooling infrastructure, air quality interventions, public health outreach, participatory planning, and anti-displacement protections all work together to strengthen both environmental and social resilience. Importantly, the project highlights that resilience is not solely about surviving climate disasters, but about creating healthier, more equitable, and more sustainable communities over the long term.

This project also underscores the importance of democratizing climate knowledge. Making environmental information accessible through community-facing educational materials and collaborative outreach efforts can empower residents to advocate for their own neighborhoods and participate meaningfully in resilience planning. Community resilience is strongest when residents are not treated as passive recipients of policy decisions, but as active partners in shaping the future of their communities.

Although this project faced limitations related to time, data access, and direct community engagement, it should be understood as a foundational and evolving contribution rather than a finalized solution. Climate resilience planning must remain adaptive, participatory, and responsive to changing environmental conditions and community needs. Future work should continue expanding direct collaboration with Germantown residents, community organizations,

and local leaders to ensure that resilience efforts remain grounded in lived experience and community priorities.

Ultimately, Germantown represents both the challenges and possibilities of equitable climate adaptation in urban communities. By centering environmental justice, hyperlocal planning, and community resilience, this project argues that climate adaptation can serve not only as a strategy for reducing environmental risk, but also as an opportunity to strengthen neighborhood stability, public health, collective care, and long-term social equity.

## Appendices:

Figures 1-3: Infographics on Rain Gardens, how to install, and resources throughout Philadelphia

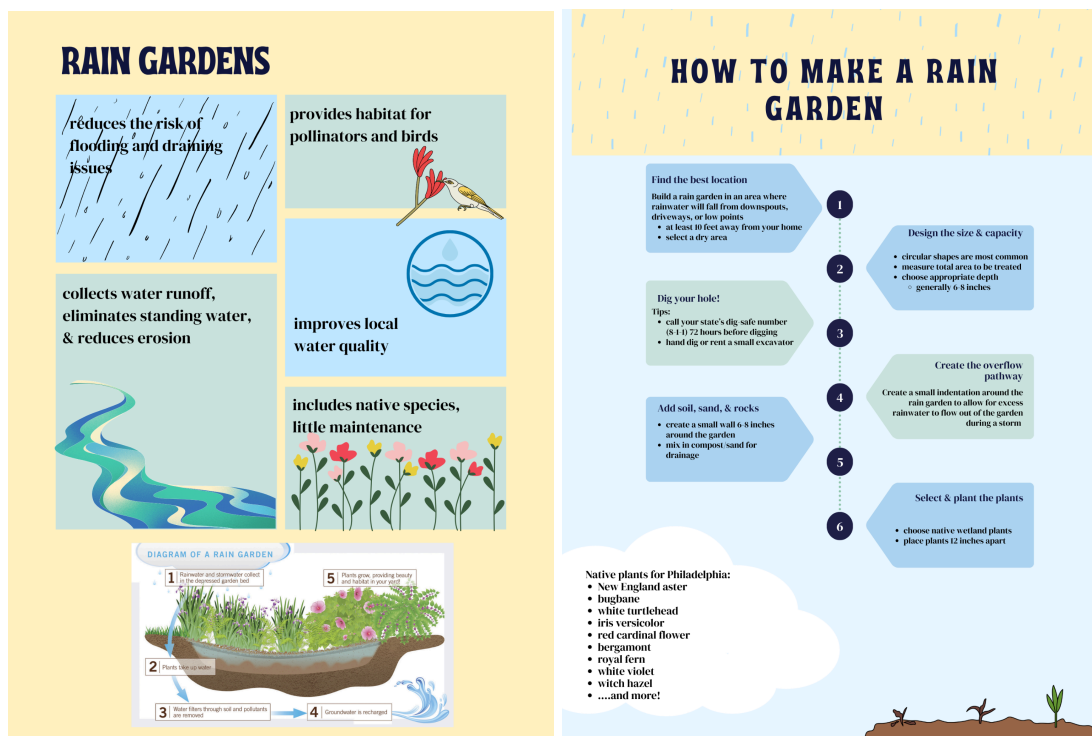


Figure 4: Infographics on the most promising USDA grants and funding resources that can be leveraged for Germantown residents and groups.

# BUILDING A STRONGER FOOD FUTURE IN GERMANTOWN

## How Germantown Residents Can Apply for USDA Grant Programs

USDA and local partners are investing in resilient food systems, supporting farmers, and improving access to healthy, local food in our communities.



### KEY USDA GRANT PROGRAMS



#### RESILIENT FOOD SYSTEMS INFRASTRUCTURE (RFSI) PROGRAM

Strengthens the "middle" of the food supply chain by funding infrastructure, equipment, and projects that improve processing, aggregation, storage, and distribution.

- Helps small and mid-sized producers reach new markets
- Supports local food businesses and supply chain resilience
- Over \$21 million in grants awarded in Pennsylvania since 2023

#### EXAMPLE IN PHILADELPHIA:

The Food Trust received RFSI funding to expand cold storage, commercial kitchen space, and transportation—bringing over 100 additional products from regional producers into Philadelphia markets.



#### FRESH FOOD FINANCING INITIATIVE (FFFI)

Invests in grocery stores, cooperatives, and food distribution businesses to improve access to fresh, nutritious food—especially in underserved communities.

- Strengthens local supply chains and food access
- Funds refrigeration, transportation, and food safety systems
- Helps address food insecurity and supports regional farmers

#### WHY IT MATTERS:

More than 1 in 9 Pennsylvanians face food insecurity. FFFI helps bring locally sourced food to neighborhoods that lack grocery stores or fresh food options.



#### PHILADELPHIA FOOD JUSTICE INITIATIVE (PFJI)

Supports community-driven solutions that advance food justice through grants to local organizations.

- Funds projects that increase access to culturally relevant, affordable, and nutritious food
- Centers equity and prioritizes historically underserved communities
- Strengthens community leadership and local solutions

#### LOCAL IMPACT:

PFJI has distributed millions of dollars to grassroots organizations working to build a more equitable and resilient food system in Philadelphia.

#### WHO CAN APPLY?

- Farmers and food producers
- Food businesses and cooperatives
- Nonprofit organizations
- Community groups and local partnerships

#### HOW TO GET STARTED

- Identify the right program for your project or organization.
- Visit [www.grants.gov](http://www.grants.gov) to find open USDA funding opportunities.
- Check eligibility requirements and application deadlines.
- Prepare a strong application that highlights community impact.
- Reach out to local partners for support!




TOGETHER, WE CAN BUILD A RESILIENT FOOD SYSTEM THAT SUPPORTS OUR HEALTH, OUR ECONOMY, AND OUR COMMUNITY.



#### LEARN MORE

Visit [www.usda.gov](http://www.usda.gov) or contact your local USDA Rural Development office for more information on grant programs and technical assistance.

Figure 5: Infographic on the fundamentals of air pollution, how it relates to Germantown, and what community-based approaches exist to address these concerns.



## Sources

- Best, K., He, Q., Flores, A. B., Gu, D., Howell, J., Johnson, D. R., Liao, Y., Reilly, A. C., Rumbach, A., Sheldon, T., Siders, A.R., Wong-Parodi, G., Young, R., & Niemeier, D. (2025).  
Bridging community resilience and local planning for climate justice. *Climate Risk Management*, 51(100776). Science Direct. doi.org/10.1016/j.crm.2025.100776
- Bullard, Robert. "Environmental Justice in the 21<sup>st</sup> Century," In John Dryzek and D. Schlosberg (eds) *Debating the Earth* (Oxford University Press, 2005).
- Bullard, Robert D. *Dumping in Dixie: Race, Class, and Environmental Quality*. 3rd ed. Boulder, CO: Westview Press, 2000.
- Center for Sustainable Systems, University of Michigan. 2025. "Environmental Justice Factsheet." Pub. No. CSS17-16.
- Chatterjee, Rhitu. "In U.S. Cities, The Health Effects Of Past Housing Discrimination Are Plain To See." NPR, November 19, 2020. Accessed May 9, 2026. NPR: Health Effects of Past Housing Discrimination.
- City of Philadelphia. "Extreme Heat Guide Resources." Accessed May 9, 2026. Philadelphia Extreme Heat Resources.
- Disruption, community, and resilient governance: Environmental justice in the Anthropocene. (2019). In T. Haller, T. Breu, T. De Moor, C. Rohr, & H. Znoj (Eds.), *The Commons in a Glocal World: Global Connections and Local Responses* (71st ed., p. 54). Taylor & Francis.

“evertreen.” 2025. Urban Reforestation: How Cities Can Combat Climate Change.

<https://www.evertreen.com/news/urban-reforestation-how-cities-can-combat-climate-change>.

Frantz, George R. 2019. “Stormwater Management Park in Your Future?” American Society of Landscape Architects.

<https://www.asla.org/news-insights/the-field/stormwater-management-park-in-your-future>

Gould, Kenneth A., and Tammy L. Lewis. *Green Gentrification: Urban Sustainability and the Struggle for Environmental Justice*. New York: Routledge, 2017.

Hansberry Garden & Nature Center – A community resource for fresh food and environmental education. Accessed May 8, 2026. <https://www.hansberrygarden.org/>.

Hearn Morrow, B. (2008). Community resilience: A social justice perspective. *Community & Regional Resilience Initiative*, 4(CARRI Research Report), 1-28.

Hidden City Philadelphia. “Repercussions of Racist Maps Still Impact Neighborhoods Today.” Accessed May 9, 2026. Hidden City Philadelphia: Repercussions of Racist Maps Still Impact Neighborhoods Today.

Historic Germantown. “Black Lives Matter Statement.” Accessed May 9, 2026. Historic Germantown: Black Lives Matter Statement.

Husband, Jessie. “Map of Germantown Philadelphia Philly.” Etsy listing. Accessed May 9, 2026. Etsy: Map of Germantown Philadelphia Philly.

Intergovernmental Panel on Climate Change (IPCC). *Climate Change 2023: Synthesis Report*. Geneva: IPCC, 2023.

- Knox-Hayes, J., Agarwal, S., et al. (2025). The Equitable Resilience Framework: An environmental justice strategy for community-led resilience planning. *World Development Perspectives*, 40(100738). ScienceDirect.
- Pearsall, Hamil, and Jillian Eller. "Locating the Green Space Paradox: A Study of Gentrification and Public Green Space Accessibility in Philadelphia, Pennsylvania." *Landscape and Urban Planning* 195 (2020): 103708.
- Philadelphia Department of Public Health. "Philadelphia Environmental Justice Screen Tool." Accessed May 9, 2026. Philadelphia Environmental Justice Screen Tool.
- Philadelphia Office of Sustainability. *Beat the Heat Hunting Park: A Community Heat Relief Plan*. Philadelphia: City of Philadelphia, 2018.
- Philadelphia Water Department. "Combined Sewer Overflow and Green Stormwater Infrastructure Programs." Accessed May 8, 2026. Philadelphia Water Department.
- Philadelphia Water Department. "Germantown Flooding Task Force." Accessed May 9, 2026. Philadelphia Water Department: Germantown Flooding Task Force.
- Philadelphia Water Department. "Wingo What?" Accessed May 9, 2026. Philadelphia Water Department: Wingo What?.
- "Philly Tree Plan." n.d. The Philadelphia City Fund.  
<https://philacityfund.org/programs/philly-tree-plan/>.
- "Rain Gardens – Green Stormwater Infrastructure." n.d. Philadelphia Water Department. Accessed May 8, 2026. <https://water.phila.gov/gsi/tools/rain-garden/>.
- Rigolon, Alessandro, et al. "Green Gentrification and Health: A Scoping Review." *International Journal of Environmental Research and Public Health* 17, no. 8 (2020): 1–26.

- Schlosberg, D. (2019). Disruption, community, and resilient governance: Environmental justice in the Anthropocene. In *The Commons in a Glocal World* (pp. 54-71). Routledge.
- Scott, Heidi. 2022. "Sustainable Agriculture Lesson: Community Gardens – Justice, Safety, and Climate Solutions." National Socio-Environmental Synthesis Center.  
<https://www.sesync.org/resources/sustainable-agriculture-community-gardens-justice-safety-and-climate-solutions>.
- "Stormwater Wetland – Green Stormwater Infrastructure." n.d. Philadelphia Water Department.  
Accessed May 8, 2026. <https://water.phila.gov/gsi/tools/wetland/>.
- Storz, E. (2024, December). *A Framework for Community-Driven Environmental Justice Guided by the Community*. Drexel News.  
<https://drexel.edu/news/archive/2024/December/framework-for-community-driven-environmental-justice>
- United States Census Bureau. "2010 Census Data for Germantown, Philadelphia." Accessed May 8, 2026. United States Census Bureau.
- United States Environmental Protection Agency. *EJScreen: Environmental Justice Screening and Mapping Tool*. Washington, DC: EPA, 2024. EPA EJScreen.
- United States Environmental Protection Agency. "Ground-Level Ozone Pollution." Accessed May 8, 2026. EPA Ground-Level Ozone Pollution.
- United States Environmental Protection Agency. "Particulate Matter (PM) Pollution." Accessed May 8, 2026. EPA Particulate Matter Pollution.
- United States Occupational Safety and Health Administration (OSHA). "Heat Injury and Illness Prevention in Outdoor and Indoor Work Settings." Accessed May 8, 2026. OSHA Heat Injury and Illness Prevention.

Visit Philadelphia. “Germantown Neighborhood Guide.” Accessed May 9, 2026. Visit Philadelphia: Germantown Neighborhood Guide.

“What Is A Climate Resiliency & Climate Adaption Plan?” n.d. Indigenous Aware.  
<https://www.indigenoussaware.com/climate-resiliency-adaptation-plan>

“What is Community Resilience?” n.d. Kansas Wesleyan University.  
<https://www.kwu.edu/crh/resources/what-is-community-resilience/>.

WHYY. “Redlining Segregated Philadelphia.” Accessed May 9, 2026. WHYY: Redlining Segregated Philadelphia.

Wikipedia contributors. “Germantown, Philadelphia.” *Wikipedia*. Last modified May 2026. Accessed May 9, 2026. Wikipedia: Germantown, Philadelphia.

Wolch, Jennifer R., Jason Byrne, and Joshua P. Newell. “Urban Green Space, Public Health, and Environmental Justice: The Challenge of Making Cities ‘Just Green Enough.’” *Landscape and Urban Planning* 125 (2014): 234–244. “4.3 Green Roofs – Development Services.” n.d. Philadelphia Water Department. Accessed April 25, 2026.  
<https://water.phila.gov/development/stormwater-plan-review/manual/chapter-4/4-3-green-roofs/>.